

DIABETES MELLITUS

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CONTENTS

Introduction	4
Classification of Diabetes Mellitus	4
Screening and Diagnosis.....	6
Screening of Asymptomatic Individuals.....	6
Diagnosis.....	8
Initial Assessment	10
Management	13
Treatment Targets.....	14
Recommended Follow-Up Interval	17
Lifestyle Modification	18
Glucose-Lowering Agents	20
Insulin Therapy.....	29
Blood Glucose Monitoring	33
Prevention of Cardiovascular Disease	38
Prevention and Management of Complications.....	40
Role of Health Team Members	48
Prevention of Type 2 Diabetes Mellitus and Managing Pre-Diabetes	51
Defining Pre-Diabetes	51
Lifestyle Intervention	51
Diet	51
Weight Management	52
Physical Activity.....	52
Smoking	52
Pharmacological Therapy.....	52
Follow-Up	53
NUP Pre-Diabetes Consult Workflow	54
Special Situations and Considerations	55
Acute Hypoglycaemia and Management.....	55
Acute Hyperglycaemia and Management	58
Female Patients at Reproductive Age	59
Gestational and Pre-Gestational Diabetes	59
Children and Adolescents with DM	60
Patients with Type 1 DM	60
Patients Fasting During Ramadan	60

Referrals.....61
Recommended Care Components for Diabetes Mellitus62
Abbreviations64
References65

INTRODUCTION

Classification of Diabetes Mellitus

Classification of Diabetes Mellitus is according to aetiological types as follows:

1. Type 1 diabetes mellitus (T1DM)
2. Type 2 diabetes mellitus (T2DM)
3. Gestational diabetes (GDM)
4. Other specific types

Type 1 Diabetes Mellitus

T1DM is characterised by β -cell destruction attributable to an autoimmune process.

Immune-mediated T1DM commonly occurs in childhood and adolescence but can occur at any age, even in the 8th and 9th decades of life. Some patients, particularly children and adolescents, may present with ketoacidosis or with an acute dramatic onset of hyperglycaemia. Adults, however, may present in a manner resembling T2DM though lacking in obesity, and with difficulty in achieving glycaemic control using oral hypoglycaemic agents, progressing quickly to an insulin-requiring state (Latent Autoimmune Diabetes of Adulthood – LADA).

Although patients are rarely obese, the presence of obesity is not incompatible with the diagnosis. These patients may also have other autoimmune disorders - such as Graves' disease, Hashimoto's thyroiditis and Addison's disease.

Type 2 Diabetes Mellitus

T2DM is characterised by disorders of insulin action and insulin secretion, either of which may be the predominant feature. Both are usually present at the time the diabetes is manifest. The specific reasons for the development of defects in insulin action and secretion are not completely elucidated.

T2DM is the most common form of diabetes. Its frequency varies in different racial/ethnic subgroups. It often has a strong familial and genetic predisposition.

The risk of developing T2DM is associated with increasing age, obesity and lack of physical activity. The risk is already significantly higher even in Singaporean Chinese with normal body mass index (BMI), rising further with increasing BMI. Women with prior gestational diabetes and individuals with hypertension and dyslipidaemia are at risk as well.

T2DM is frequently undiagnosed for many years because hyperglycaemia develops gradually and at earlier stages, it is often not severe enough for the patient to notice any of the classic symptoms of diabetes. Nevertheless, such patients are at increased risk of developing

macrovascular and microvascular complications. These individuals initially, and often throughout their lifetimes, do not need insulin for survival. However, people with T2DM may require insulin treatment at some stage of their disease. Ketoacidosis is infrequent and usually arises in association with the stress of another illness such as infection.

Gestational Diabetes Mellitus

GDM is defined as the onset or first recognition of any degree of glucose intolerance during pregnancy. This category does not include women with glucose intolerance that predates the pregnancy, whether already known to have diabetes or previously unrecognised and discovered only during the first antenatal visit. The definition applies irrespective of whether insulin is used for treatment, or whether the condition persists after pregnancy.

The diagnosis and management of GDM, and management of women with diabetes and pregnancy, are detailed in the [Gestational and Pre-Gestational Diabetes](#) section.

Other Specific Types

This refers to those relatively uncommon causes of diabetes in which the underlying defect or disease process can be specifically identified.

These include genetic defects of β -cell function or insulin action, diseases of the exocrine pancreas (pancreatitis, pancreatectomy) as well as diabetes induced by other endocrinopathies (Cushing's syndrome, acromegaly, glucagonoma, phaeochromocytoma, hyperthyroidism), drugs (nicotinic acid, glucocorticoids, thiazides, interferon, diazoxide), toxins and infections (congenital rubella, cytomegalovirus). Monogenetic defects of β -cell function, of which several subtypes exist, are characterised by early age of onset of hyperglycaemia, strong family history of diabetes displaying autosomal dominant pattern of inheritance, absence of autoimmunity and usually low insulin requirement with relative ease in achieving glycaemic control with sulphonylureas (Maturity Onset Diabetes of the Young – MODY).

SCREENING AND DIAGNOSIS

Screening of Asymptomatic Individuals

Screening should be considered in adults of any age who have one or more risk factors for diabetes. In those without risk factors, testing should begin at 40 years.

Subsequently, screening should be carried out every three years for those with normal HbA1c or glucose tolerance and annually for those with impaired fasting glycaemia (IFG) or impaired glucose tolerance (IGT).

Risk factors for diabetes include:

- Overweight / obesity (Asian body mass index $> 23.0 \text{ kg/m}^2$)
- First degree relative with diabetes mellitus
- High risk race / ethnicity (e.g. Black African, African Caribbean and South Asian)
- Women who have delivered a baby 4 kg or more; or previously diagnosed with gestational diabetes mellitus
- Hypertension ($> 140/90 \text{ mmHg}$) or on therapy for hypertension
- HDL cholesterol level $< 1.0 \text{ mmol/L}$ (male), $< 1.3 \text{ mmol/L}$ (female) and/or triglyceride level $> 2.2 \text{ mmol/L}$
- Women with polycystic ovarian syndrome
- History of cardiovascular disease

Screening may be performed using HbA1c, FPG, or OGTT. **Glucometers** should **not** be used for the diagnosis of diabetes mellitus.

Other Specific Types

The criteria and actions for screening with HbA1c are summarised in the table below:

Table 1: Interpretation and Action for screening of Diabetes using HbA1c and follow-up tests Results

Result	Interpretation	Action	Further Result	Recommendation
HbA1c 6.0% or lower	Low probability of DM	No further test needed. If clinical suspicion is high, consider co-existence of conditions that may affect reliability of HbA1c (see below) and conduct further investigations as clinically indicated		No Diabetes Screen at recommended interval
HbA1c 6.1 to 6.9%	Probability of DM or pre-DM	Proceed to do fasting plasma glucose or OGTT	FPG 6 mmol/L or lower OR 2H-OGTT < 7.8 mmol/L	
			FPG 6.1 to 6.9 mmol/L OR 2H-OGTT 7.8 to 11.0 mmol/L	Prediabetes Diagnose and manage as pre-DM (see page 20)
			FPG 7.0 mmol/L or higher OR 2H-OGTT 11.1 mmol/L or higher	Diabetes Diagnose and manage as DM
HbA1c 7% or higher	High probability of DM	Diagnose and manage as DM		

When using HbA1c to screen for and/or diagnose DM, clinicians must be aware of the following factors other than blood glucose which affect the HbA1c level:

- Haemoglobin variants (such as HbE variant) can interfere with some laboratory methods, particularly the cation-exchange HPLC method. By contrast, immunoassays are generally not susceptible to interference by haemoglobin variants that are common in our local population. Thalassemia is the most common genetic blood disorder locally, estimated to affect 4% of the population. A local study has found that 2.3% of blood samples from diabetes patients contain haemoglobin variants.
- Other conditions that could potentially affect red cell turnover. These include sickle cell disease, pregnancy (second and third trimesters), iron/ vitamin B12 deficiency, chronic liver disease, reticulocytosis, recent blood loss or transfusion, and erythropoietin therapy. G6PD deficiency is also common in Singapore and the impact of genetic variants in our population on HbA1c is currently being assessed.
- Situations that affect protein glycation. These include alcoholism, chronic renal failure, abnormal intra-erythrocyte pH, aspirin, vitamin C and E, certain haemoglobinopathies and genetic variants.

Diagnosis

In patients with hyperglycaemic crisis, diabetes mellitus can be diagnosed without further testing.

In patients with typical symptoms, diabetes mellitus can be diagnosed if any one of the following is present:

1. Random plasma glucose* ≥ 11.1 mmol/L
2. Fasting plasma glucose (FPG)** ≥ 7.0 mmol/L
3. 2-hour post-challenge plasma glucose ≥ 11.1 mmol/L (avoid performing OGTT on patients with typical symptoms)

Other individuals should have a repeat test on a subsequent day.

Individuals with FPG from 6.1 to 6.9 mmol/L should undergo a 75 g oral glucose tolerance test to determine if they have impaired glucose tolerance (IGT) or diabetes mellitus. Oral glucose tolerance test (OGTT) - 75 g glucose, is performed in accordance with WHO recommendations. The 1-hour post-challenge glucose is not useful for the diagnosis of diabetes mellitus.

Glucometers should **not** be used for the diagnosis of diabetes mellitus.

When two different tests are available for the same patient and the results for both tests are above the diagnostic thresholds, the diagnosis of diabetes is confirmed.

When two different tests are available in an individual and the results are discordant, the test whose result is above the diagnostic cut point (usually the fasting plasma glucose or 2-hour post-challenge glucose) should be repeated.

If a second test fails to confirm the diagnosis, barring a laboratory error, such patients are likely to have test results near the margins of the threshold for a diagnosis. The healthcare professional might opt to **follow the patient closely and repeat the testing in 6 - 12 months**.

* *Random plasma glucose refers to plasma glucose at any time of the day, without regard to the interval since the last meal.*

** *Fasting is defined as no consumption of food or beverage except water for at least 8 hours.*

Other Categories of Glucose Tolerance

Intermediate states of glucose metabolism termed impaired fasting glucose and impaired glucose tolerance should be recognised as defined in the table below:

Table 2: Intermediate Categories of Glucose Tolerance

	Fasting plasma glucose (mmol/L)		2-hour plasma glucose (mmol/L)
Impaired Fasting Glycaemia (IFG)	6.1 - 6.9	and	< 7.8
Impaired Glucose Tolerance (IGT)	< 7.0	and	7.8 - 11.0

These two intermediate categories represent an increased risk for the development of DM and cardiovascular disease (see section on [Prevention of Type 2 Diabetes Mellitus and Managing Pre-Diabetes](#)).

Initial Assessment

Initial assessment comprises evaluation of the following using history, physical examination and where relevant investigation(s):

- a) Diagnosis
- b) Co-morbid conditions (such as presence of hypertension or dyslipidaemia in a patient with DM)
- c) Complications
- d) Target organ disease
- e) Medication
- f) Others, e.g. psycho-social-economic screening where relevant

Diagnosis

- a) Determine if Type 1 or Type 2 DM
- b) Evaluate duration and severity of medical condition
- c) Evaluate if symptomatic
- d) Most patients have primary DM, hypertension or dyslipidaemia. Very rarely, there is a need to exclude secondary causes if the patient has medical history which predisposes the patient to secondary DM, hypertension or dyslipidaemia

Examples of secondary causes of DM are:

Endocrine: Cushing's syndrome, thyrotoxicosis, acromegaly

Drugs: Long term oral steroid ingestion, diuretics

Complications

Evaluate if the following complications are present:

- a) Diabetic retinopathy
- b) Diabetic nephropathy
- c) Diabetes foot complications - dermopathy, peripheral neuropathy
- d) Atherosclerotic cardiovascular diseases

Medication

Determine all medication that patient is taking on long term basis, regardless of whether related to DM, hypertension, dyslipidaemia or not.

Others (see PM-CS-AOP-008 and WI-CS-AOP-008 General Screening / Patient Assessment)

Assess:

- a) Psycho-socio-economic screening where relevant
- b) Learning Needs / Barriers (during initial assessment or by second consultation within 3 months)

Summary Table of Assessment 1

	DOCTOR / APN ##	EPIC	CARE MANAGER
Initial Assessment (At Diagnosis)	Explain diagnosis Assess risk factors Start dietary control Order DM Panel Test (add screening ECG as baseline if none previously) Order DRP Order DFS Refer care manager for disease education KIV refer dietitian Consider oral anti-diabetic agent if BGL are very high (e.g. FBS > 10 mmol/L, random glucose > 15 mmol/L). Metformin should be initiated as first line therapy for patients requiring OHGA if eGFR is > 45ml/min. For patients whose HbA1c at diagnosis is > 9%, consider initiating dual OHGA (agents should be from 2 separate drug class) treatment For patients with thirst, polyuria and polydipsia, severe uncontrolled DM HbA1c > 10% or random glucose > 16.7mmol/L, insulin therapy can be considered for initial treatment	Diagnosis Smoking, BMI	Check BMI and BP Assess learning needs Discuss DM and diet Discuss effects of medication / Hypoglycaemia (If needed, discuss and teach insulin injection / SBGM) Advise on diet / exercise Stop smoking, discourage alcohol KIV make appointment for DM workshop
Reassessment (Second Visit) Usually 2 to 6 weeks' post-diagnosis	Review patient's understanding of DM Review FBG and HbA1c +/- Drug therapy Examine feet KIV direct ophthalmoscopy Examine at diagnosis if patient has complaints Refer orthopaedic surgeon, vascular surgeon or podiatrist if necessary Encourage type and frequency of exercise suitable for patient.	DRP DFS	Review capillary glucose results / dietary intake Complications of DM Hypo-/ hyperglycaemic symptoms Sick Day Management Foot and skin care Set goal(s) on lifestyle modification and treatment
Reassessment (Third Visit) Usually 2 – 3 months after previous visit	Review symptoms (e.g. hypoglycaemia / hyperglycaemia, side effects with medicine, etc.) Review HbA1c +/- FBG or 2HPP +/- Drug therapy Review DM Panel Test Review DRP and DFS findings Refer ophthalmologist, dietitian if necessary	DM Panel Test comprising: HbA1c Lipid Panel Serum Creatinine U, ACR	Review patient progress Review goal(s) Re-educate patient according to their needs
Reassessment (Subsequent Visits) # / ## Every 3 - 4 months (may vary from 2 - 6 months)	Review symptoms Review HbA1c, BP, BMI.	HbA1c BP BMI	Review patient progress Review goal(s) Re-educate patient according to their needs
Annual Assessment ## (Yearly)	Review level of control, presence of risk factors, co-morbid conditions (such as hypertension and/or dyslipidaemia) and DM complications. Manage accordingly where necessary (e.g. initiation / titration of medication, referral to dietitian, podiatrist, psychologist, MSW, hospital SOC)	DM Panel Test DRP DFS	Review patient progress Review goal(s) Re-educate patient according to their needs Foot and skin care

Note

The intervals of visits depend on the condition of the patient. Patients with high FBS (e.g. > 10 mmol/L) or RBS (e.g. > 13 mmol/L), who are very symptomatic but without sign / symptom of HHS may require earlier review within 2 weeks after initial diagnosis.

Summary Table of Assessment 2

PARAMETER	DISEASE / CO-MORBIDITY / COMPLICATIONS	FREQUENCY OF REASSESSMENT / ANNUAL ASSESSMENT
HbA _{1c}	Glycaemic control	HbA_{1c} (3-6 monthly) * FBG / RBG (prn)
Weight	Overweight / obesity	Weight (every 2-6 monthly)
BP	Blood pressure control	BP** (every 2-6 monthly)
Others	Peripheral Neuropathy Peripheral vascular disease Dyslipidaemia IHD Nephropathy Retinopathy	Examine distal extremities (yearly)*** DM panel test (yearly) ECG (baseline done once upon diagnosis) U, ACR (yearly) and S, creatinine / eGFR DRP (yearly) or Eye review at hospital dept.
Smoking	Cerebrovascular disease	Advise to stop smoking at each visit
Note: * Check HbA_{1c} 3 - 4 monthly if poor glycaemic control or if there is recent adjustment or therapy or intensive insulin therapy. ** Check for postural drop in BP at initial assessment and periodically especially if there is neuropathy. The frequency of monitoring of weight and BP monitoring depends on the condition of the patient (e.g. a patient with BP > 180/110 mmHg will need to be treated and reviewed within the same day). Refer to CPGs on Hypertension and Obesity. *** Examine feet at each visit if there is gross neuropathy or PVD, monitor lipids, serum creatinine more frequently if abnormal. # The Clinical Pharmacist will be involved in the patient care based on WI-PHA-PFE-003 Clinical Pharmacist Clinic – Hypertension, Diabetes, Lipids (CPC – HDL). Where alternate consultation is made between Dr and Care Manager / Nurse Clinician, tests such as DM Panel test, HbA_{1c}, lipid profile, DRP and DFS may be ordered by the Care Manager. Refer NUP-PM-NUR-AOP-005 Care Management, NUP-WI-NUR-AOP-005 Chronic Disease Management by Care Manager and NUP-PM-CS-COP-011 Laboratory Investigations. ## The APN will be involved in the patient care based on NUP-PM-NUR-AOP-001 Advanced Practice Nurse. APN will re-direct patient to polyclinic Dr if referral to hospital SOC is needed and discuss case with polyclinic Dr if referral to Allied Healthcare Professional is needed.		

MANAGEMENT

Explain the medical condition, assess individualised coronary risk factors and initiate dietary modification.

- 1) Blood glucose review
 - a) Review the patient's parameters such as FBG or HbA1c, BP, BMI.
 - b) Decide, based on the blood glucose result, whether dietary control alone is sufficient to maintain glycaemic control.
 - c) Drug therapy is generally initiated at a low dose with slow titration of doses according to glycaemic control at subsequent consultation during reassessment if needed.
- 2) Incidence of therapeutic failure increases with duration of DM. Doctors should:
 - a) Check patient's compliance with medications and dietary recommendation.
 - b) Exclude on-going sepsis including dental source.
 - c) Encourage healthy lifestyle, e.g.
 - exercises appropriate for patient's age and medical conditions
 - weight reduction in overweight / obese patients
 - smoking cessation among smokers

Consider insulin therapy if HbA1c is persistently above 8% and patient is already on maximal doses of oral anti-hyperglycaemic agents. Earlier initiation of medication is to be considered if the patient:

- a) is symptomatic (e.g. presence of polydipsia, polyuria, significant weight loss) with deranged glucose levels despite dietary modification (e.g. RBS > 16.7 mmol/L or FBS > 10 mmol/L).
- b) has recurrent symptomatic complications of hyperglycaemia such as candidiasis.

Doctors should order DM panel test, DRP and DFS generally within the first or second consultation for patients with DM upon diagnosis and subsequently annually. DRP and DFS may be ordered by nurses and other members of the care team as well.

The doctor may perform direct ophthalmoscopy if the patient refuses DRP. The doctor may also perform foot screening if the patient is not keen to have DFS. The dates of "manual" eye and foot examination are to be recorded in the patient's notes.

The doctor may refer the patient with persistently poor glycaemic, BP or lipid control to:

- a) Second Tier Clinics if available. Second Tier Clinics are run by senior doctors (e.g. Family Physicians or Resident Family Doctors) to help optimise patients' medical conditions. Where these clinics are not available, doctors may discuss management plans with senior doctors during consultation.
- b) Other health team members (See section on Role of Health Team Members)

Insulin use should be considered in patients with extreme high levels of glycaemia; HbA1c > 10% or random glucose > 16.7 mmol/L.

Rarely, patients with Type 1 DM may present in the polyclinic. These patients are generally young and have acute onset of symptoms with marked weight loss.

- a) If the clinical presentation is suggestive of impending Diabetic Ketoacidosis (DKA), the Type 1 DM patient should be hydrated rapidly with intravenous normal saline. Intravenous insulin should be given and patient sent via ambulance to the A&E department.
- b) If the T1DM patient is stable, he can be referred via Direct Access to the specialist outpatient clinic after initial treatment by the polyclinic doctor.

Medical Officers should consult family physicians to review patients with the following features:

- a) BGL $\geq$ 20 mmol/L.
- b) presence of clinical features suspicious of a diabetic emergency [hyperosmolar hyperglycaemia state (HHS) and DKA], e.g., fever, drowsiness, vomiting, abdominal pain, dehydration.
- c) presence of symptoms suggesting severity of hyperglycaemia such as significant loss of weight or significant polydipsia.

Treatment Targets

Monitoring of glycaemic status is regarded as a cornerstone of diabetes care. It provides information to help physicians and patients make adjustments to diabetes treatment. Improved glycaemic control is associated with sustained decreased rates of retinopathy, nephropathy and neuropathy. However, patients striving to achieve the best level of glucose control they can achieve may encounter hypoglycaemia. Therefore, glycaemic targets must be individualised to ensure that patients do not incur an undue risk of hypoglycaemia (see table below).

Two methods are available to access glycaemic control: patient self-monitoring of blood glucose (SMBG) and glycated haemoglobin (HbA1c).

Table 4. Target for Glycaemic Control

Test	Targets	For frail patient susceptible to hypoglycaemia ^{##}
HbA _{1c} (%)	< 7%	7.0 - 8.5%
Pre Meal Glucose (mmol/L)	4.0 - 7.0	6.5 - 9.0
2HPP (mmol/L)	5 - 10	12

^{##} Sinclair AJ, Paolisso G, Castro M et al. Rodriguez Manˆas L; European Diabetes Working Party for Older People. European Diabetes Working Party for Older People 2011 clinical guidelines for type 2 diabetes mellitus: Executive summary. *Diabetes Metab* 2011;37(Suppl. 3):S27–S38.

Glycated Haemoglobin

Glycated haemoglobin (HbA1c) should be performed routinely in all patients with diabetes, at initial assessment and then as part of follow-up care. The measurement of HbA1c quantifies average glycaemia over the previous 2 - 3 months, thereby complementing blood glucose testing which provides information on day-to-day glycaemic excursions.

The HbA1c test is subject to certain interferences. **Haemoglobin variants** and **conditions that affect red blood cell turnover** (e.g., blood loss, haemolysis) may affect the validity of HbA1c results. These conditions should be considered particularly when the HbA1c results do not correlate with the patient's clinical condition.

A small number of patients with Hb variant would have invalid HbA1c results due to unacceptable imprecision despite using alternative methods (DC100 machine). They should have their glycaemic control assessed with regular (self) [blood glucose monitoring or continuous blood glucose monitoring](#). These readings should be interpreted alongside clinical findings (symptoms, presence / absence of complications) to provide an overall picture of glycaemic control.

The **frequency of HbA1c** testing should be determined by the clinical situation, the treatment regimen used, and the judgement of the clinician. The following schedule is recommended for HbA1c testing in patients with diabetes:

- 3- to 4-monthly in patients with unstable glycaemic control, failure to meet treatment goals, recent adjustment in therapy, or intensive insulin therapy
- 6-monthly in patients who have stable glycaemic control and who are meeting treatment goals

A HbA1c target of $\leq 7.0\%$ for most non-pregnant adults with type 1 or type 2 diabetes has been shown to reduce the risk of developing microvascular complications and if implemented soon after the diagnosis of diabetes, is associated with lowered risk of macrovascular disease.

Lowering HbA1c target to $\leq 6.5\%$ may be considered for some patients with T2DM, if this can be achieved without significant hypoglycaemia. Such patients include those with short duration of diabetes, long life expectancy and no significant cardiovascular complications.

Less stringent HbA1c target (e.g. 7.0 to 8.5%) may be adopted for some patients vulnerable to the harmful effects associated with tight glycaemic control. Such patients include those with very long duration of diabetes, known history of severe hypoglycaemia, advanced atherosclerosis and advanced age. Doctors should be vigilant in preventing hypoglycaemia by reviewing treatment regimens in patients with near-normal HbA1c levels (e.g. $< 6.0\%$), especially those treated with insulin or insulin secretagogues.

Cholesterol

(See CPG on [Lipids](#)).

Blood Pressure

(See section on [Prevention of Cardiovascular Disease](#)).

Body Mass Index

Keep BMI < 25 kg/m²

For Asian population, aim for BMI < 23 kg/m².

Recommended Follow-Up Interval

Criteria	Recommended TCU Frequency	Recommended for alternate Dr / CM visit
• Good glycaemic, blood pressure, lipid level control, AND • Not on insulin, AND • Does not have complication	6 months	√
• Good glycaemic, blood pressure, lipid level control, AND • Elderly or has complications such as IHD, CVA, renal impairment	3 - 4 months	√
• Compliant with follow-up and treatment, AND • With or without co-morbidities / complications, AND • Stable but sub-optimal control related to individual targets for BP, HbA1c and Cholesterol over the past 4 -6 months	3 - 4 months	To discuss with Teamlet CM
• Started on insulin ± HBGM	1 - 8 weeks	Consider CM insulin titration, see Section on <u>Insulin Therapy</u> for details
• With adjustment to medication	6 weeks - 3 months	√

Lifestyle Modification

Lifestyle modification is a cornerstone of diabetes management and includes the following:

- Medical nutrition therapy
- Physical activity and exercise
- Avoidance of smoking

Medical Nutrition Therapy

Medical nutrition therapy in diabetes addresses not only glycaemic control but other aspects of metabolic status as well, including dyslipidaemia and hypertension - major risk factors for cardiovascular disease.

Individuals who have diabetes should receive individualised medical nutritional therapy as needed to achieve treatment goals, preferably provided by a dietitian familiar with the components of diabetes medical nutrition therapy.

Consider referring newly diagnosed (duration < 12 months) motivated individuals to the NUH Diabetes Remission Clinic. The 6-month program offers personalised care supported by tailored nutrition and smart technology, aim at significantly reducing the risk of diabetes-related complications. Providers may refer to the [patient brochure](#) (with FormSG sign up QR code) and [estimated costs](#). Patients expressing interest will be contacted by NUH for further advice and financial counselling.

Meal and snack carbohydrate intake for diabetes should be consistently distributed throughout the day, on a day-to-day basis, as consistency in carbohydrate intake has been shown to result in improved glycaemic control. (Grade B, Level 2+, 1)

If weight reduction is needed, it should be attempted gradually (0.25 to 1.0 kg/week). In overweight or obese patients with type 2 diabetes, a weight loss of 5-10% of body weight achieved through lifestyle interventions is a realistic goal. (Grade B, Level 2+, 1)

Recommendations for fibre intake for people with diabetes are similar to the recommendation for the general population. A daily consumption of a diet containing 20-35 g of dietary fibre from a wide variety of food sources is recommended. (Grade B, Level 2+, 1)

Minimize or avoid sugary beverages entirely. Assess and address alcohol use disorder. Heavy alcohol consumption is linked with elevated HbA1c and increased risks of hypoglycaemia especially in patients on insulin or sulphonylureas.

Individuals who choose to use non-nutritive sweeteners should be advised that some of these products might contain energy and carbohydrate from sources that might need to be accounted for. (Grade B, Level 2, 1)

Physical Activity and Exercise

Exercise in T2DM leads to improvement in glycaemic control, contributes to weight loss and has a favourable effect on cardiovascular risk factors.

Patients with T2DM should undertake at least 150 mins/week of moderate to vigorous aerobic exercise spread out during at least 3 days of the week, with no more than 2 consecutive days between bouts of exercise.

An exercise programme should be tailored for each individual patient to suit his age, aptitude, fitness and interest. Before embarking on a programme more vigorous than brisk walking, a pre-exercise evaluation is important especially in sedentary and older diabetic individuals to identify any cardiovascular disease, hypertension, neuropathy or microvascular retinopathy changes that might contraindicate certain types of exercise and predispose to injury.

There is no definitive evidence to suggest that pre-exercise ECG stress testing for asymptomatic individuals at low risk of cardiovascular disease is beneficial but it may be indicated for higher risk individuals.

Exercise may help prevent the onset of early peripheral neuropathy. Patients with peripheral neuropathy and without acute ulceration may participate in moderate weight-bearing exercise. Comprehensive foot care, use of appropriate footwear and daily foot check is recommended.

Exercise and physical activity generally do not adversely affect vision or the progression of non-proliferative diabetic retinopathy or macular disease. However, patients with severe proliferative diabetic retinopathy should avoid activities that greatly increase intraocular pressure and risk of haemorrhage such as weightlifting, heavy Valsalva manoeuvres, and heavy competitive sports may be contraindicated as these may trigger vitreous haemorrhage and retinal detachment.

In patients on insulin treatment or secretagogues, physical activity can cause hypoglycaemia if medication dose or carbohydrate consumption is not altered. Certain precautions can avoid hypoglycaemia; these include monitoring of blood glucose levels before and after exercise, adjustment of insulin and secretagogues prior to exercise or consumption of extra carbohydrate prior to or during exercise if exercise is unplanned and glucose level is low.

Avoidance of Smoking

Assessment of smoking status as well as smoking cessation counselling and support should be a routine component of diabetes care. Patients with diabetes should be encouraged to stop smoking. Nicotine replacement therapy is subsidized (MAF claimable) and can be offered at pharmacist led smoking cessation clinics.

Glucose-Lowering Agents

When lifestyle intervention (i.e. healthy diet, physical activity, weight control) is insufficient in achieving glycaemic control, initiate glucose-lowering agents whilst continuing lifestyle intervention.

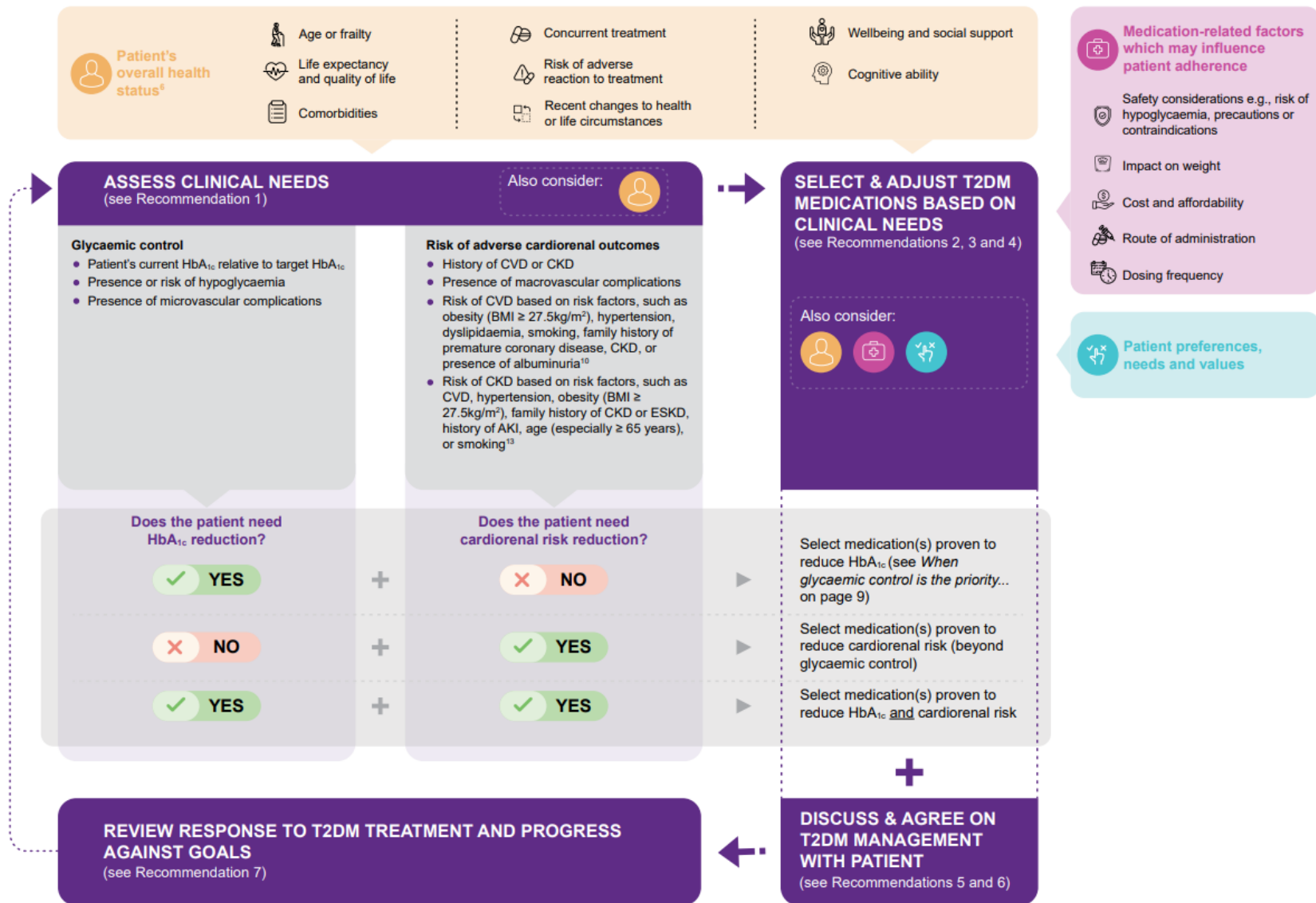
Oral agents are preferred over injections due to convenience and acceptability. However, **do not delay insulin therapy in patients with symptomatic hyperglycaemia, unexplained recent weight loss irrespective of initial weight, ketonuria or diabetic ketoacidosis.**

As the duration of diabetes increases, insulin resistance develops and pancreatic β -cell function progressively declines. Multiple glucose-lowering agents are required to achieve and maintain glycaemic control. Combining agents with different mechanisms of action and side effects can enhance glycaemic control and tolerability.

Insulin therapy should be considered if optimal combination therapy fails to attain target control (i.e. 2 consecutive HbA1c values failed to reach $\leq 8\%$ over 3 - 6 months interval).

In the therapeutic journey, the progressive nature of T2DM and treatment should be explained and reinforced regularly, and progression to insulin therapy should not be described as failure or punishment.

Figure 4 provides an algorithm for use of glucose-lowering agents. (From Personalising management with non-insulin medications. Agency for Effective Care, MOH)



AKI, acute kidney injury; BMI, body mass index; CKD, chronic kidney disease; CVD, cardiovascular disease; ESKD, end-stage kidney disease; HbA_{1c}, glycated haemoglobin. Points listed in figure provide examples of factors to be considered; it is not an exhaustive list.

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Metformin (MET) - Preferred Initial Glucose-Lowering Agent				
Metformin Immediate Release (IR) (S1) (250 mg and 500 mg Tablets) \$	500 mg OM	750 mg TDS	Nausea, anorexia, vomiting, diarrhoea, abdominal cramps, cholestatic hepatitis. Risk of lactic acidosis in patients with renal or hepatic insufficiency or severe cardiovascular disease.	Metformin has good HbA1c lowering (1 - 1.5%) and long-term safety data. Titrate the dose gradually and advise patients to take metformin with or after meals to reduce gastrointestinal side effects, such as nausea, vomiting and diarrhoea.
Metformin Immediate Release (IR) (S1) (850 mg Tablets) \$	850 mg BD	850 mg TDS		Use metformin with caution in patients with renal impairment and those at risk of lactic acidosis (e.g. patients with hepatic impairment or heart failure). Consider discontinuing Metformin for the duration of acute illness in those with eGFR 30 - 59 who experience illnesses such as diarrhoea and vomiting.
Metformin Extended Release (XR) (NS) (500 mg Tablet) \$\$\$\$	500 mg OM	2000 mg OM		Long-term use of metformin may be associated with vitamin B12 deficiency (< 220 pmol/L). In patients with established anaemia or peripheral neuropathy, monitor B12 levels as part of routine care.

Table 6. Dosing Recommendation for Metformin in Renal Impaired Patients

eGFR (ml/min)	Recommendation
> 60	Use normal dosing
45 - 59	Use normal dosing, monitor renal function more closely (3-6 monthly)
30 - 44	Maximum dose of 500 mg BD (or Metformin XR 1g OM)
< 30	Stop Metformin

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Sodium-glucose cotransporter-2 (SGLT-2) inhibitors - For Patients who are at Risk of Hypoglycaemia, are Overweight, or with Cardiovascular Disease				
Dapagliflozin (Forxiga®) (NS) (10 mg Tablets) \$\$\$\$	5 mg or 10 mg once daily	10 mg once daily	Thirst, increased urination	Sodium-glucose cotransporter-2 (SGLT-2) inhibitors lower HbA1c (0.6 - 0.9%), body weight and blood pressure, and may benefit those who are overweight or obese. They have been shown to improve cardiovascular outcomes in T2DM patients. Avoid SGLT-2 inhibitors in patients at risk of DKA (e.g. suspected T1DM, severe hyperglycaemia, recent surgery, infection, low caloric or carbohydrate intake, or pancreatic insufficiency). SGLT-2 inhibitors work by blocking renal glucose reabsorption and are not recommended for patients with severe renal impairment (Table 8). Patients on SGLT-2 inhibitors should also observe good genital hygiene. Always counsel patients to seek urgent medical attention if they have symptoms of DKA and pain in their lower limbs. Consider them as a class and choose the lowest cost option. SGLT-2 inhibitors have been associated with an increased risk of urinary and genital tract infections, euglycemic diabetic ketoacidosis (EDKA) and lower limb amputations (Canagliflozin).
Empagliflozin (Jardiance®) (S2) (10 mg and 25 mg Tablets) \$\$	10 mg or 25 mg once daily	25 mg once daily		
Canagliflozin (Invokana®) (100 mg and 300 mg Tablets) (Not available in NUP)	100 mg or 300 mg once daily	300 mg once daily		

^ Cost \$ <\$10 per month \$\$\$\$ \$30 - <\$60 per month
 \$\$ \$10 - <\$20 per month \$\$\$\$\$ \$60 - <\$90 per month
 \$\$\$ \$20 - <\$30 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Table 8. Starting Dosing Recommendations for SGLT-2 Inhibitors in Renal Impairment

Estimated Glomerular Filtration Rate (mL/min/1.73m ²)	Dapagliflozin (Forxiga)	Empagliflozin (Jardiance)	Canagliflozin (Invokana)
≥ 60	10 mg once daily	10 mg or 25 mg once daily	100 mg or 300 mg once daily
≥ 30 to < 60	10 mg once daily at eGFR ≥ 45 ml/min. Initiation for glycaemic control alone not recommended at eGFR < 45 ml/min but shown to have cardiorenal benefits up to initiation of dialysis and can be continued.	10 mg or 25 mg once daily	100 mg once daily
< 30		Initiation for glycaemic control alone not recommended at eGFR < 30 ml/min but shown to have cardiorenal benefits up to 20 ml/min and can be continued at 10 mg once daily.	Not recommended

MAF subsidy for dapagliflozin ceased in August 2024 and for patient with cost concerns, clinicians can discuss switching to empagliflozin which is subsidised (SDL2).

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Glucagon-like peptide 1 (GLP-1) receptor agonists – For Patients who are Overweight and Improves Atherosclerotic CVD (ASCVD) Outcomes				
Dulaglutide (Trulicity®)* (0.75/1.5 Subcutaneous injection)	0.75 mg weekly (Titrate to 1.5 mg weekly after 1 month if tolerating well)	1.5 mg weekly	Predominantly gastrointestinal particularly nausea, vomiting and diarrhoea. Gradual dose titration helps, consider as needed symptomatic therapy (e.g., proton pump inhibitor). Be more cautious in patients with pre-existing renal disease at risk of acute kidney injury from volume depletion.	Glucagon-like peptide 1 (GLP-1) receptor agonists reduce HbA1c (1 -1.5 %) and achieve significant weight loss (up to 9.6 kg) by augmenting glucose dependent insulin secretion and slowing gastric emptying. These agents have also been shown to improve cardiorenal outcomes. When used in combination with basal insulin, result in less hypoglycaemia or weight gain while achieving glycaemic targets with lower insulin doses. Should NOT be used together with DPP-4 inhibitors as no additive effects in glucose lowering. Counsel patients: - side effects usually transient - small meals, eat slowly - stop before feeling full - avoid fatty, spicy, strong flavoured foods
Semaglutide (Wegovy®) (0.25/0.5/1/1.7/2.4 mg Subcutaneous injection) Indicated for patients with diabetes and BMI ≥ 27 kg/m ² (Not available in NUP)	SC 0.25 mg weekly	SC 2.4 mg weekly	Small risk of hypoglycaemia when started in conjunction with insulin, sulphonylureas or meglitinides. Consider concomitant dose reduction.	Contraindicated in patients with a personal or family history of medullary thyroid carcinoma (MTC); patients with multiple endocrine neoplasia syndrome type 2 (MEN2); pregnancy (requires washout of at least 6 weeks when trying to conceive) Caution in patients with history of pancreatitis, possible gastroparesis, and irritable bowel disease. Consider pancreatitis and stop GLP-1 immediately in patients with suggestive symptoms, i.e. unremitting epigastric pain that radiates to back with nausea and vomiting. Refer newly initiated patients to CM for injection technique.

* GLP-1 receptor agonists should be initiated or endorsed by senior doctors in the Teamlets.

The Ministry of Health's Drug Advisory Committee has recommended Dulaglutide 0.75 mg and 1.5 mg solution for injection in pre-filled pen for treating type 2 diabetes mellitus:

- as a triple therapy in combination with two oral anti-diabetic drug (OAD) therapies for patients with inadequate glycaemic control despite treatment with optimal doses of dual OAD therapy, or as a dual therapy in combination with one OAD therapy if a dual OAD therapy is contraindicated or not tolerated; and
- in combination with insulin and metformin, unless metformin is contraindicated or not tolerated.

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Sulfonylureas (SU) – Use Second Generation Sulfonylureas				
Glipizide (Minidiab) (S1) (5 mg Tablets) \$	2.5 mg OM	15 mg BD	Very infrequently mild nausea, vomiting, diarrhoea, allergic reactions.	Sulfonylureas lower HbA1c by 1 - 1.5%, are affordable and have long-term safety data.
Gliclazide (S1) (80 mg Tablets) \$	40 mg OM	160 mg BD		However, sulfonylureas cause weight gain and pose a risk of hypoglycaemia, especially in the elderly and those with renal or hepatic impairment.
Gliclazide MR (Diamicon) (NS) (60 mg Tablets) \$\$\$	30 mg OM	120 mg OM		Use second generation sulfonylureas, such as glipizide and gliclazide, which are less likely to cause hypoglycaemia as compared to other sulfonylureas. Avoid glibenclamide which have higher hypoglycaemia risks.
Tolbutamide (S1) (500 mg Tablets) \$	250 mg BD	1000 mg TDS		Modified-release (MR) gliclazide and glimepiride are taken once daily and may improve patient adherence. However, these tablets are costlier compared to glipizide or immediate-release (IR) gliclazide.
Glibenclamide (Daonil) (5 mg Tablets) (Not available in NUP)	2.5 mg OM	10 mg BD		Do not use glibenclamide in older patients aged ≥ 60 years old or patients with renal impairment (eGFR < 60 , Creatinine above normal Limit) to avoid risk of hypoglycaemia. Switch to other short-acting sulphonylureas such as glipizide or gliclazide.

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Dipeptidyl peptidase-4 (DPP-4) inhibitors - For Patients with Renal Impairment				
Linagliptin (Trajenta®) (NS) (5 mg Tablets) \$\$\$ MAF eligible in patients with T2DM and CKD $\geq$ 3B	5 mg once daily	5 mg once daily		Dipeptidyl peptidase-4 (DPP-4) inhibitors have similar HbA1c lowering (0.5 - 0.8%) as SGLT-2 inhibitors. They are weight neutral, well tolerated and rarely cause hypoglycaemia. Local cost-effectiveness analyses have shown that MET + DPP-4 inhibitor is not cost-effective compared to MET + SU or MET + SGLT-2 inhibitor. However, consider DPP-4 inhibitors for patients unable to use sulfonylureas (i.e., hypoglycaemia or weight gain) or SGLT-2 inhibitors (i.e., renal or other side effects). DPP-4 inhibitors are suitable for patients with renal impairment with dose adjustments as shown in Table 9. Linagliptin does not require dose adjustments for patients with declining renal functions.
Sitagliptin (Januvia®) (S2) (50 mg and 100 mg Tablets) \$	50 or 100 mg once daily	100 mg once daily		

Table 9. Dosing Recommendations for DPP-4 Inhibitors in Renal Impairment

Estimated Glomerular Filtration Rate (mL/min/1.73m ²)	Linagliptin (Trajenta)	Sitagliptin (Januvia)
≥ 45	5 mg once daily	100 mg once daily
≥ 30 to < 45	No dose adjustment needed	50 mg once daily
$< 30^{\dagger}$	No dose adjustment needed	25 mg once daily

[†] Including end-stage renal disease on haemodialysis

Long-term cardiovascular outcome trials suggest no significant differences in composite cardiovascular endpoints. However, saxagliptin and alogliptin may increase heart failure hospitalisations, particularly in patients with heart or renal disease. The United States Food and Drug Administration (FDA) has issued safety warnings on heart failure with the use of these drugs.

Other Glucose-Lowering Agents

Other oral glucose-lowering agents include meglitinides, thiazolidinediones and alphaglucoSIDase inhibitors. Their safety and cost limit their widespread use.

Using fixed-dose combinations, such as a sulfonylurea, SGLT-2 inhibitor or DPP-4 inhibitor with metformin, reduce pill burden and improve patient adherence. However, they are less flexible for dose titration.

Drug	Initial Dose	Maximum Dose	Common ADR	Remarks / Contraindications / Precautions
Alpha-Glucosidase Inhibitors				
Acarbose (Glucobay®) (NS) (50 mg and 100 mg Tablets) \$\$\$\$	50 mg TDS	100 mg TDS	Flatulence, soft stools, diarrhoea – usually dose-related. Small risk of hypoglycaemia if taken with insulin and sulphonylurea.	
Thiazolidinedione				
Rosiglitazone (Avandia®) (4 mg Tablets) (Not available in NUP)	4 mg OM	8 mg OM	Rosiglitazone is contraindicated in patients with IHD, CCF and not recommended in patients with PAD.	Restricted access programme has been implemented for Avandia since July 2011.
Pioglitazone (Actos) (15 mg and 30 mg Tablets) (Not available in NUP)	15 mg OM	45 mg OM		Pioglitazone is associated with increased bladder cancer risk and must be avoided in those with active bladder cancer or those with history of bladder cancer.

^ Cost \$ <\$10 per month \$\$\$\$ \$30 - <\$60 per month
 \$\$ \$10 - <\$20 per month \$\$\$\$\$ \$60 - <\$90 per month
 \$\$\$ \$20 - <\$30 per month \$\$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Insulin Therapy

Insulin replacement therapy is required for all patients with T1DM and for many with T2DM.

Insulin therapy should be considered if optimal combination therapy fails to attain target control (i.e., **2 consecutive HbA1c values failed to reach $\leq 8\%$ over 3 - 6 months interval**).

In the therapeutic journey, the progressive nature of T2DM and treatment should be explained and reinforced regularly, and **progression to insulin therapy should not be described as failure or punishment. Many patients with uncontrolled diabetes feel better once started on insulin** (more energy, better sleep, less urinary frequency, improved mood).

The key to effective insulin therapy is an understanding of principles that, when implemented, can result in improved diabetes control; unfortunately, tight glycaemic control is associated with a certain degree of risk of hypoglycaemia.

Patients on insulin must be equipped not only with the skills of insulin administration, but also should be educated on self-monitoring of blood glucose, hypoglycaemia management, matching of insulin dose and carbohydrate intake, and dose adjustments during sick days, travel, exercise, and changes in food intake.

Initiation and Intensification of Basal Insulin Therapy

This may be initiated as a bedtime dose of intermediate-acting or long-acting insulin with maintenance of oral agents during the day.

Intensification to Multi-Dose Insulin Therapy

When glycaemic control is not achieved despite the addition of basal insulin to oral agents, discontinuing sulphonylureas and switching to premixed twice daily or basal-bolus insulin regimens becomes necessary. However, metformin, SGLT-2 inhibitors, DPP-4 inhibitors and α -glucosidase inhibitors may still be used in conjunction with exogenous insulin to attenuate the insulin dose.

Fine-tuning of insulin doses is best determined by home blood glucose monitoring.

Patients with T2DM who are switched to insulin therapy temporarily during episodes of acute stress, such as sepsis, may be put back on oral agents when their glycaemic control improves with declining insulin resistance and glucotoxicity.

CLINICAL SITUATION	MANAGEMENT	WHAT TO DO?
Type 2 DM patients with therapeutic failure on oral medications	Start basal insulin	DOCTOR : * For insulin initiation: Start with 8-10 units or (0.1unit/kg BW) units intermediate-acting or long-acting insulin at 10pm or OM (morning). For insulin titration by CM: with additional 4 weeks' duration and state clearly in eRx remarks: CM Insulin Titration. Start from XX units to YY units, e.g., CM to titrate from 8U to 16U as per protocol. Order Dr with lab (e.g. HbA1c on arrival) ± CM visit in 8-12 weeks (based on clinical judgement). NURSE : Teach injection technique, HBGM, hypo- / hyperglycaemic symptoms. REVIEW : CM to do a telephone call to patient within 1 to 4 days to follow up whether patient has started on insulin injection, injecting the correct dosage, any coping problems and any symptoms of hypoglycaemia. CM to review patient within 1-2 weeks / Check injection sites. Review injection technique. TITRATE : Increase insulin by 2-4 units (or 10% of total dose) at each visit / through CM weekly phone consult until adequate glycaemic control is achieved. For basal insulin given at night, titrate insulin dosing using morning fasting glucose. Consider insulin regimen intensification [1. Basal plus (add second injection of fast acting pre-biggest meal), 2. Premix b.d. or 3. Basal bolus] in the following situations: a) if morning glucose is at target even if HbA1C remains high, b) if basal insulin dosage is at threshold level of 0.5 unit/kg BW.
Inadequate glycaemic control while on oral medications + basal insulin with good fasting pre-breakfast sugar levels	Consider adding on short-acting insulin (basal plus regimen) Consider converting to twice a day pre-mixed insulin regime	DOCTOR : Gradually taper off sulphonylurea. i. Add short-acting insulin (6U or 0.1unit/kg BW) with main meal of the day ii. Convert to twice a day pre-mixed insulin (e.g., Mixtard 30/70, Humalog Mix25) Titrate gradually in 2-4 units increments until adequate glycaemic control is achieved (pre next meal and/or post meal sugars). Consider CM tele-titration support as described above Add insulin based on following general guidelines: • Patient may continue taking biguanide, SGLT-2, DPP-4 while on intensified insulin. • Total daily insulin requirement for Type 2 Diabetes patients = 0.5-2 unit/kg ideal body weight

CLINICAL SITUATION	MANAGEMENT	WHAT TO DO?
Switching patient from pre-mixed insulin regime to basal plus or basal bolus	Convert to 70% (or 75%) basal insulin and 30% (or 25%) prandial insulin. ^For patients with higher risk of hypoglycaemia, consider more conservative dose adjustment: Reduce total daily basal insulin dose by 20%	Example of conversion: Current therapy – Humalog Mix25 20 units pre-breakfast and 10 units pre-dinner DOCTOR : <u>Suggested dose of basal insulin</u> = 75% of total daily dose (for Humalog Mix25) = $0.75 \times (20 + 10)$ = E.g., Lantus 22 units once daily (rounded down) ^ For patients with higher risk of hypoglycaemia, reduce total daily dose by 20% E.g., $0.8 \times 0.75 \times (20 + 10) = 18$ units once daily (rounded down) <u>Suggested dose of prandial insulin for basal plus</u> = 25% of dose given with the heaviest meal (for Humalog Mix25) = 0.25×20 units = E.g., Apidra 5 units with heaviest meal of the day <u>Suggested dose of prandial insulin for basal bolus</u> = 25% of total daily dose (for Humalog Mix25) and divided across 2 – 3 meals = $0.25 \times (20 + 10)$ = 7.5 units E.g., Apidra 3 units with lunch and 4 units with dinner NURSE : Review injection technique, HBGM, and educate on hypo- / hyperglycaemic symptoms.

Note:

* Consult senior doctor if you have never started insulin therapy on your own. Consider referral to specialist if patient's insulin requirement is > 1 unit/kg ideal body weight.

Examples of Types of Insulin

	Classification	Vial (1000 units in 10ml)	Pen (300 units in 3ml)	Flexpen (300 units in 3ml)
a)	Long-acting		Insulin Glargine (Lantus®)	
b)	Intermediate-acting	NPH Insulin (Insulatard®)		
c)	Short-acting	Soluble Insulin (Actrapid®)	Insulin Glulisine (Apidra®)	Insulin Aspart (NovoRapid®)
d)	Pre-mixed	Mixtard 30 HM (30% Soluble Insulin / 70% NPH Insulin)		Humalog Mix25 (25% Insulin Lispo / 75% Insulin Lispro Protamine) Novomix 30 (30% Aspart / 70% Aspart Protamine)

Patients should have a meal within 30 minutes of injecting short-acting insulin (including pre-mixed) for regular insulin and 15 minutes for insulin analogues.

Blood Glucose Monitoring

Self-monitoring of blood glucose (SMBG) is recommended for patients with type 1 or type 2 diabetes who are treated with insulin. SMBG can guide adjustment of insulin and allows patients to assess whether glycaemic targets are being achieved. Results of SMBG can also be useful in preventing hypoglycaemia in this group of patients.

SMBG should be considered in the following groups of patients with type 2 diabetes who are not treated with insulin:

- Patients at increased risk of developing hypoglycaemia or its consequences (e.g., patients who are using sulphonylureas)
- Pregnant patients with pre-existing diabetes or gestational diabetes
- Patients experiencing acute illness
- Patients who have failed to achieve glycaemic goals
- Patients undergoing fasting, for example, during Ramadan

The frequency and timing of SMBG should be determined by the particular needs and goals of the patients. For example:

- Pre breakfast fasting sugars in titration of basal insulin
- 2-hour post meal sugars in patients embarking on dietary changes
- Prior to and during strenuous exercise / operating heavy machinery / driving while on insulin or sulphonylureas

For patients with T1DM, a retrospective cohort study showed that mean HbA1c was significantly decreased in patients who carried out SMBG at least 3 times a day compared to those who performed an average of 1 SMBG a day.

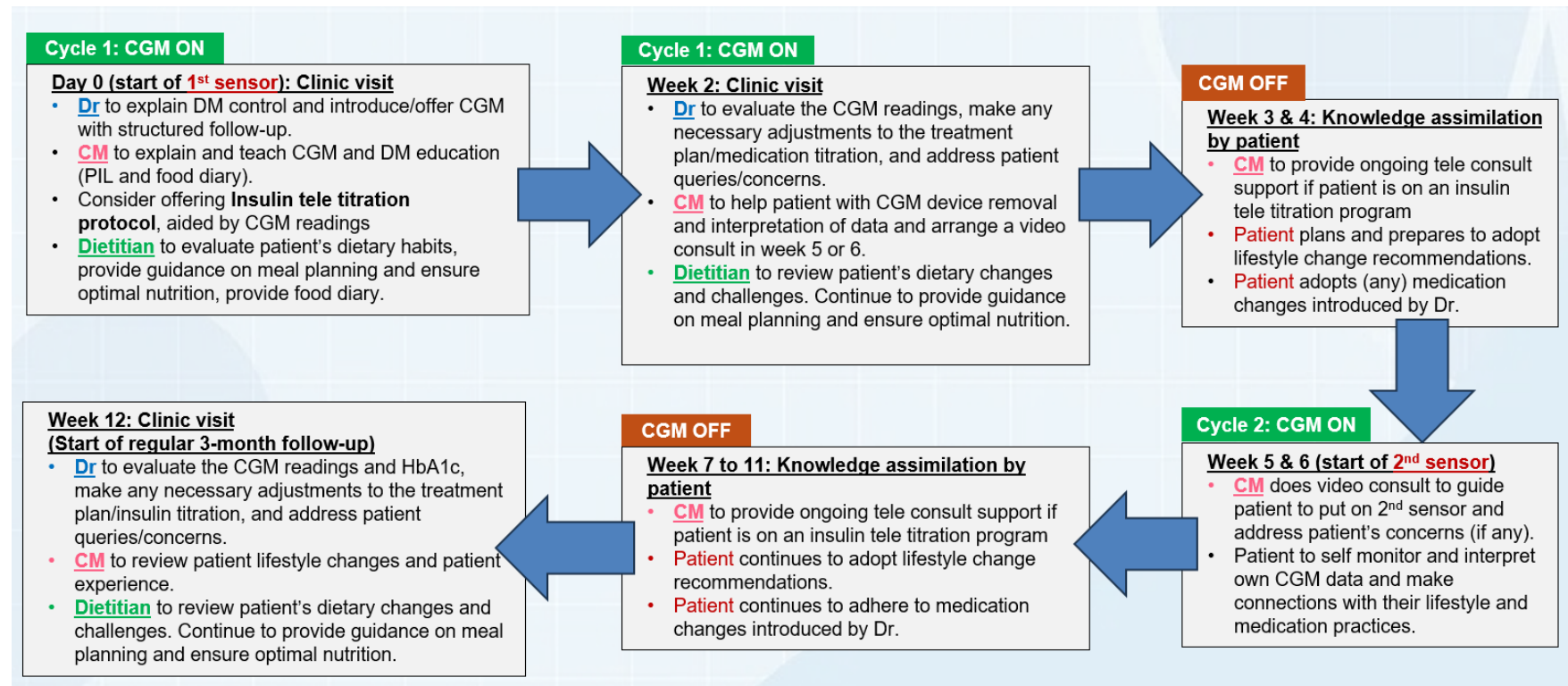
For intensive insulin-treated T2DM patients, testing 2 or 3 times a day on 2 to 3 days a week would be appropriate.

For patients with unstable metabolic control, changes in daily routine, alterations of treatment regimens or acute illness, the frequency of self-monitoring of blood glucose should be increased.

SMBG in isolation does not reduce blood glucose levels and needs to be integrated into a structured clinical and self-management program.

Intermittently Scanned (isCGM) and **Real Time Continuous blood glucose monitoring (rtCGM)** are new forms of monitoring technology that have shown to be beneficial in patients with T1DM or T2DM, especially patients on insulin therapy. They are also useful short-term tools for troubleshooting in patients who experience great difficulty in managing their condition.

Consider enrolling patients newly prescribed with continuous glucose monitoring on a structured disease education program. The following workflow was successfully piloted with encouraging outcomes as part of DM Collaborative and PTEC.



Interpreting an Ambulatory Glucose Profile

First, determine if action is needed by looking at page showing summary metrics. General goals to keep in mind

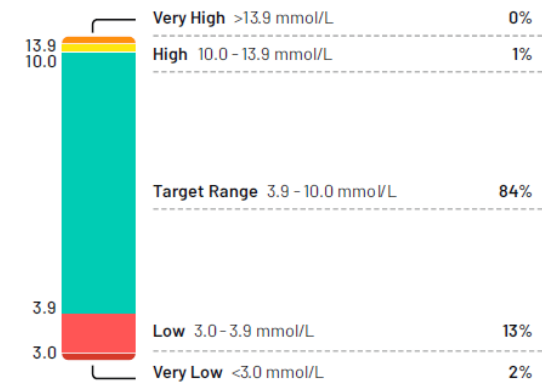
- a) Data Quality: At least 14 days of data with > 70% wear time for reliability
- b) Time In Range (TIR): > 70%
- c) Time Below Range (TBR): < 4% (with <1% in the very low range < 3.0 mmol/L)
- d) Time Above Range (TAR): < 25%
- e) Glucose Variability: $\leq 36\%$

Glucose Statistics and Targets

15/10/2025 - 28/10/2025 14 days
% Time Sensor Active 96%

Ranges and Targets for	Type 1 or Type 2 Diabetes
Glucose Ranges	Targets % of Readings (Time/Day)
Target Range 3.9 - 10.0 mmol/L	Greater than 70% (26h 49min)
Below 3.9 mmol/L	Greater than 4% (47h 16min)
Below 3.0 mmol/L	Greater than 1% (6h 11min)
Above 10.0 mmol/L	Less than 25% (3h 53min)
Above 13.9 mmol/L	Less than 5% (0min)
Average Glucose	5.5 mmol/L
Glucose Management Indicator (GMI)	5.7% $GMI(\%) = 3.31 + 0.427 \times \text{Mean Glucose (mmol/L)}$
Glucose Variability	31%
Defined as percent coefficient of variation (%CV); target $\leq 36\%$	

Time in Ranges



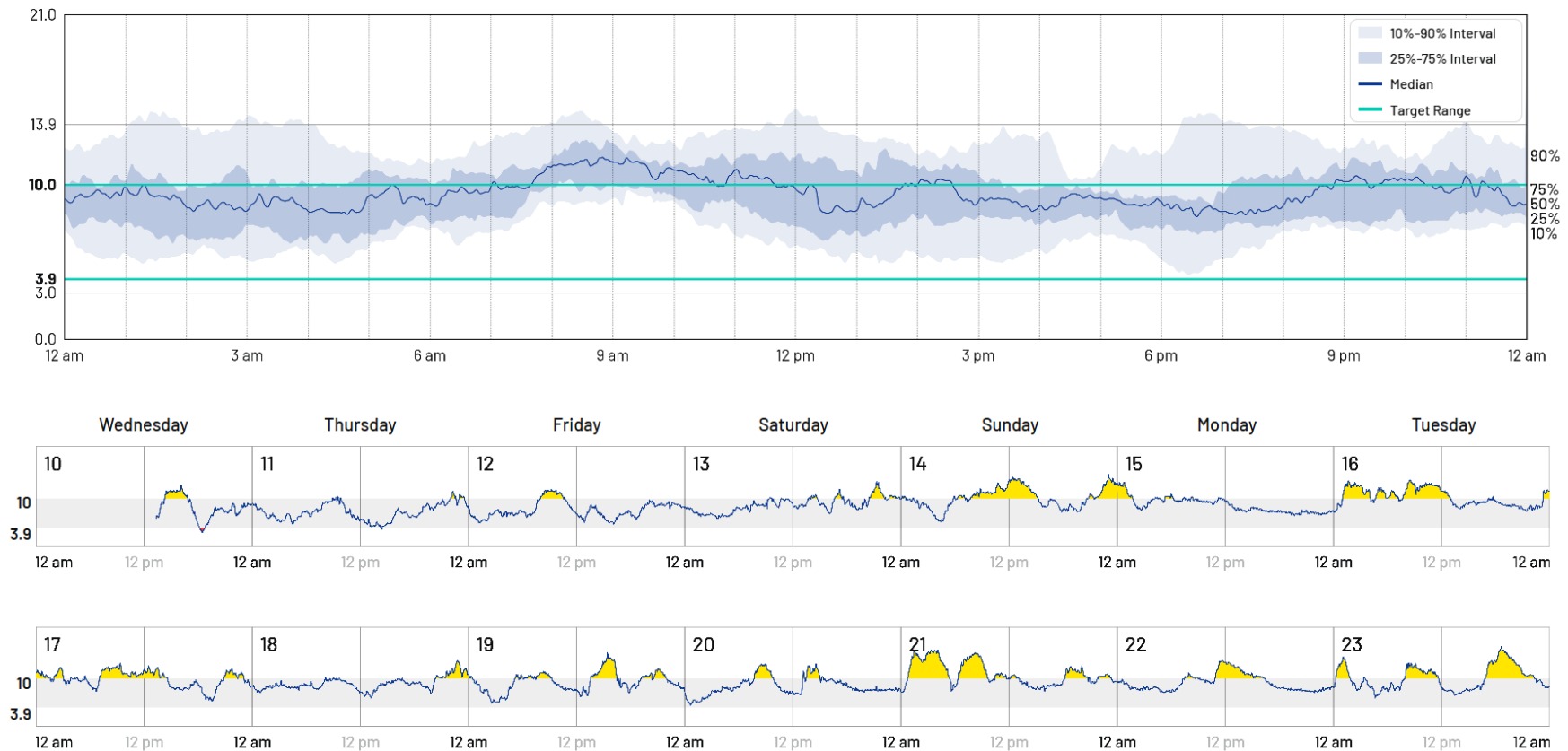
In this summary, the patient has good data quality and TIR but significant TBR (15%) concerning for frequent hypoglycaemia

Next refer to the ambulatory glucose profile and individual daily glucose profiles to assess

- What time of the day are glucose values low < 4.0 mmol/L or high > 10.0 mmol/L
- Whether there are certain days of the week that glucose values tend to not be in range

Ambulatory Glucose Profile (AGP)

AGP is a summary of glucose values from the report period, with median (50%) and other percentiles shown as if occurring in a single day.



In this patient, the glucose spikes late morning and middle of night were due to excessive carbohydrate intake from his only 2 meals a day as a private hire driver

Finally act on the data to improve time in target glucose range:

- a) Address hypoglycaemia in patients on insulin or sulphonylureas
- b) Ask patient about
 - i) Diet (heavy meals precipitating glucose spikes, types of food)
 - ii) Activity (skipping meals, strenuous exercise, missed readings if any while on isCGM)
 - iii) Medication compliance
- c) Address hyperglycaemia
 - i) Consider SGLT2 inhibitors or GLP1 receptor agonists
 - ii) Titrate basal or bolus insulin depending on whether fasting or post meal glucose values are elevated
- d) Consider tele-consult in 2 to 4 weeks for follow-up contact especially if insulin was titrated or GLP1-RA started

Urine glucose monitoring

Self-monitoring of urine glucose is not recommended for monitoring of glycaemic status.

Prevention of Cardiovascular Disease

Global cardiovascular assessment

People with diabetes are at higher risk of developing cardiovascular disease than people without diabetes and diabetes has sometimes been regarded as a coronary risk equivalent. Identifying the varying cardiovascular risk factor burden is an integral part of the management of the patient with T2DM.

The assessment of cardiovascular risk in persons with type 2 diabetes mellitus should include:

History - which should include:

- Smoking
- Hypertension
- Pre-existing cardiovascular disease (including angina, myocardial infarction, stroke, PVD)
- Family history of premature coronary artery disease (non-modifiable)

Physical examination - which should include:

- Assessment for peripheral vascular disease
- Measurement of blood pressure at every visit

Tests - which should include:

- Lipid Panel at or soon after diagnosis and at least annually thereafter
- Urine albumin or protein at least annually
- Serum creatinine and estimation of eGFR
- Electrocardiogram (resting) routinely at baseline. Subsequent ECG may be performed when clinically indicated.

Measures to prevent cardiovascular disease in patients with T2DM

There is compelling evidence that, in general, multiple risk factor management reduces cardiovascular mortality in patients with T2DM.

Management of blood glucose in patients with T2DM with a view to minimising cardiovascular risk

Intensive blood glucose lowering in people with newly diagnosed diabetes may have potential cardiovascular benefits and these benefits may accrue many years later. However, these benefits may not be accrued by intensive glucose lowering to below 6.5% in patients with long-standing T2DM.

Management of Hypertension in Patients with T2DM

The presence of hypertension in patients with T2DM is associated with increased risk of microvascular complications (retinopathy and nephropathy) as well as macrovascular complications (stroke, CHD, and peripheral vascular disease).

For patients with T2DM who have hypertension, an acceptable treatment-initiation and target blood pressure is < 130/80 mmHg.

ACE inhibitors and ARBs have been shown in earlier trials to be beneficial in terms of reducing incident microalbuminuria and as well as in retarding progression of diabetic nephropathy. An angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) should be included as part of antihypertensive regimen for patients with T2DM requiring pharmacotherapy for hypertension, unless not well tolerated.

The choice of antihypertensive agents in people with diabetes should also look into whether there are compelling indications or possible contraindications.

Management of Dyslipidaemia in Patients with T2DM

(See Lipids CPG).

Management of the Prothrombotic State in Patients with T2DM

T2DM is a prothrombotic state which may predispose affected individuals to cardiovascular disease.

Though effective in reducing cardiovascular morbidity and mortality in secondary prevention, the role of aspirin in primary prevention in patients with T2DM is controversial. It may be considered after a comprehensive discussion on the benefits versus comparable risk of bleeding with patients aged ≥ 50 years old especially in the concomitant presence of 1 additional major risk factor (family history of premature atherosclerotic cardiovascular disease, hypertension, hyperlipidaemia, smoking or chronic kidney disease). The risk of bleeding is especially higher in patients over the age of 70 years.

Prevention and Management of Complications

Diabetes Foot Screening (DFS)

In Singapore, there is an average of four lower extremity amputations (LEA) a day in people with diabetes. About 3 in 4 LEA are preceded by diabetic foot ulcers (DFU). In addition to LEA, DFU are associated with mobility loss, poorer quality of life, and decreased overall productivity. Regular foot assessment is recommended to identify and manage DFU risk. This guide outlines key considerations in conducting foot assessment to identify and manage DFU risk.

1. Assessment of the feet in patients with diabetes should begin with checking for active or urgent diabetic foot conditions. If present, patients should be managed accordingly (see table 11).
2. In the absence of active or urgent diabetic foot conditions, patients should be assessed for symptoms and/or signs of callus, deformity, peripheral arterial disease (PAD), and neuropathy.
3. If symptoms and/or signs of the above are present, patients should be referred to the appropriate specialty (see table 12).
4. Based on the medical history and symptoms and/or signs, the patient should be assigned a risk category. The frequency of repeat assessment is based on the patient's risk category at each assessment (see Table 13).
5. Foot examination and tests can be performed by doctors in the consultation room (see Table 14).

Table 11 - Management of Active or Urgent Diabetic Foot Conditions

Presentation	Management
• Signs of moderate or severe infection and/or sepsis • Acute Charcot arthropathy, such as redness, swelling, or warmth • Cellulitis or pus from wound • Wet gangrene	• Refer to Emergency Department
• Tissue loss (ulcer or blister) • Rest pain • Absent pulses / $ABI \leq 0.9$ / absolute toe pressure of < 60 mmHg • Mild or localised infection/poor healing of any lower limb ulcer • Dry gangrene	• Initiate treatment as appropriate e.g. Epic Order "Podiatry, Foot Attack" under Reason for referral: Expected date <i>within 3 days</i> • Refer to NUH Vascular Hot Clinic via Direct Access • Consider antiplatelet therapy in symptomatic peripheral artery disease and treat LDL to target < 2.1 mmol/L

Table 12 - Foot Assessment and Referral Decisions

	Medical History	Symptoms and Examination			
History and Examination	History of previous foot ulcer or lower extremity amputation OR CKD stage 5 (eGFR < 15 ml/min/1.73m ²)	Pain or functional impairment, and/or Deformity, including • Charcot arthropathy • Hallux valgus (bunion) • Hammer or claw toe if it increases rubbing in footwear, or pressure	Rest pain, vascular claudication, and/or severe or worsening symptoms of neuropathy, with findings suggestive of PAD, and/or Absence of any pedal pulse with either ABPI ≤ 0.9 or toe pressure < 60 mmHg	Severe or worsening symptoms, such as numbness, pain, or tingling sensation, without findings suggestive of PAD and Inability to feel 10 g monofilament at any of the four tested sites on each foot. OR Inability to feel vibration fully from activated 128 Hz tuning fork, as assessed by examiner. OR Inability to feel vibration from neurothesiometer probe at ≤ 25 V (average).	Callus, including: • Simple callus • Thick callus requiring treatment • Callus with intradermal bleeding
Presume or Diagnose as	High Risk	Deformity	Peripheral Arterial Disease (PAD)	Neuropathy	Callus
Refer to	NUP DFS Nurse	Orthopaedics	Vascular or NUP DFS Nurse	Neurology	NUP Podiatry
Referral Notes	See Table 13 on Risk stratification and Frequency of Assessment		See Table 11 on Management of active or urgent diabetic foot conditions Patient with vascular claudication should be referred to vascular as a routine referral. In the absence of active or urgent diabetic foot conditions or vascular claudication, refer patient to the NUP DFS nurse for foot assessment and lifestyle modification e.g. physical activity, smoking cessation, <i>etc.</i>	Features suggesting atypical forms of diabetic neuropathy or causes of neuropathy other than diabetes, such as sudden onset, asymmetrical distribution, or motor deficits worse than sensory deficits should also be referred to Neurology	See Table 13 on Risk stratification • High Risk patients Order "Podiatry High Risk": <i>seen within 2 to 4 weeks</i> • Moderate or Low Risk patients Order "Podiatry First Visit with Reason for Referral" – "Podiatry, Painful callus or corn" – "Podiatry, Pathological toenails" – "Podiatry, Others, PLS SPECIFY" <i>seen within 2 to 3 months</i>
Remarks	The patient need not see the DFS Nurse if he/she is on regular follow-up with the NUP Podiatrist.	Charcot arthropathy is suggested by clinical or X-ray findings, such as mid-foot collapse, joint destruction, joint subluxation, or malalignment	ABPI >0.9 in people with diabetes does not exclude PAD, as it could be a false elevation		

Table 13 - Risk Stratification and Frequency of Assessment

Presentation / Diagnoses or Presumed Diagnoses	None OR Simple callus	Thick callus requiring treatment OR Deformity with simple callus or thick callus requiring treatment OR One of the following: • Deformity • PAD • Neuropathy	Previous foot ulcer or amputation OR CKD stage 5 OR Callus with intradermal bleeding OR Two or more of the following: • Deformity • PAD • Neuropathy
Risk Level	Low	Moderate	High
Frequency of Assessment	At least once a year	At least once every six months	At least once every three or four months

Table 14 – Foot Examination and Tests that can be performed by Doctors

Deformity	Peripheral Arterial Disease	Neuropathy
Inspect both feet thoroughly by examining skin integrity, especially areas with callus. Look for deformity (e.g. hammer toe, hallux valgus) that increases rubbing in footwear or pressure.	Palpate pedal pulses (dorsalis pedis and posterior tibial) on each foot.	Ask patient to close both eyes. Place an activated 128 Hz tuning fork over the interphalangeal joint of each big toe. Ask patient to inform examiner when vibration is no longer felt.

Routine Vaccinations

Offer all patients with diabetes updated influenza vaccinations at regular intervals and pneumococcal vaccine, especially those aged 65 years and above.

Diabetic Neuropathy

Diabetic peripheral neuropathy is the most common complication of diabetes mellitus, affecting up to 50% of patients with diabetes. Diabetic neuropathy can present as a glove-and-stocking distribution of numbness, tingling or weakness that may be painful and can spread proximally. Many patients are asymptomatic, highlighting the importance of regular diabetic foot screening.

Glycaemic, blood pressure and lipid management remain the primary mechanisms of preventing diabetic peripheral neuropathy.

Diabetic neuropathy is a diagnosis of exclusion and other causes should be considered and ruled out first. These may include hypothyroidism, toxic neuropathy from alcohol misuse, medications such as previous chemotherapy or amiodarone, and infections like Hepatitis C. Assessment for vitamin B deficiency may be useful for individuals at risk of vitamin B deficiency (e.g., long-term metformin use, strict vegetarian diets).

Management

2. Communicate clear treatment goals and optimise glycaemic, blood pressure and lipid control.
3. Non-pharmacological treatment options include:
 - a) Exercise (aerobic, resistance and balance exercises have all been shown to have modest benefit)
 - b) Several neuro-modulatory strategies have demonstrated benefit (e.g., spinal cord stimulation, transcutaneous electrical nerve stimulation)
4. It is **not** recommended to prescribe neuropathic pain agents for painless neuropathy.
5. For painful diabetic neuropathy, pharmacological options include:

Drug	Initial dose	Suggested maximum dose	Common ADR	Remarks / Contraindications / Precautions
Gabapentinoids				
Pregabalin (S2) \$\$ 25mg and 75mg capsule	75 mg – 150 mg (in 2 – 3 divided doses)	300 – 600 mg (in 2 – 3 divided doses)	Blurred vision, dizziness, dry mouth, fatigue, headache	Renal adjustment needed from <60ml/min
Gabapentin (S2) \$\$ 100mg and 300mg capsule	300 mg ON	1200 – 3600 mg (in 2 – 3 divided doses)	Ataxia, dizziness, drowsiness, fatigue, peripheral oedema	Renal adjustment needed from <80ml/min
Antidepressants				
Duloxetine (Not available in NUP) 30mg and 60mg capsule	30 mg OD (KIV uptitrate to 60 mg OD after 1-2 weeks)	60 mg OD	Decreased appetite, drowsiness, dry mouth, fatigue, nausea, vomiting	

Drug	Initial dose	Suggested maximum dose	Common ADR	Remarks / Contraindications / Precautions
Amitriptyline (S1) \$ 10mg and 25mg tablet	10-25 mg at bedtime	50 – 150 mg at bedtime or in 2 divided doses	Confusion, constipation, giddiness, dry mouth, fatigue, somnolence, urinary retention	May cause dose-dependent ECG changes (e.g., QT prolongation, etc.) Useful in concomitant depression, anxiety, or chronic migraine.
Topicals				
Topical Lidocaine (5%) Patch (Not available in NUP)	Apply up to 12 hours a day		Application-site burning, erythema, irritation, vesicles	

6. Avoid using chronic opiate medications to treat painful diabetic neuropathy.

Referral Considerations

Consider referral to neurologist if patient's symptoms do not respond to 1st line options above.

Diabetic Nephropathy (see also NUP CPG on [Chronic Kidney Disease](#))

Diabetic nephropathy is an important cause of chronic kidney disease (CKD). Diabetic nephropathy is also associated with substantial increase risk for cardiovascular disease (CVD). The chief strategy to prevent and delay its onset and progression is **good glycaemic control, and blood pressure control**.

In early diabetic nephropathy, there is micro-albuminuria (30 - 299 mg albumin/g creatinine or 30 - 299 mg albumin/24h). Beyond this, it is termed macro-albuminuria.

For patients with proteinuria, a target blood pressure of 130/80 mmHg appears reasonable.

Screening

1. Screen UACR (freshly collected sample) and Serum Creatinine at least annually for patients with no existing nephropathy or kidney diseases.
2. For patients with existing nephropathy or kidney diseases
3. For patients with heavy albuminuria (≥ 300 mg/g), measurement of urinary protein/creatinine ratio (PCR) should be used.

Management

1. **Good glycaemic control** is the most important factor for delaying progression. It is also the prerequisite for the other measures to be effective.
2. Advise patient to stop smoking.
3. Treat dyslipidaemia (see [Lipids](#) CPG).
4. Monitor UACR / UPCR; Serum Creatinine and electrolytes 3-monthly to yearly (depending on severity of renal impairment).
5. In patients with T2DM, hypertension, and microalbuminuria, treatment with ACE-I or ARBs is recommended.
6. SGLT-2 inhibitors reduce the risk of kidney disease progression in diabetic nephropathy and is recommended.
7. In patients with T2DM, hypertension, macroalbuminuria, and renal insufficiency (serum creatinine > 130 µmol/L), ARBs are recommended. ACEI, ARB should be used cautiously in people with eGFR < 30 ml/min.
8. If serum creatinine rises > 30% from baseline with initiation or increase in dose of ACE-I or ARB, stop the medication or reduce medication to previous dose after excluding other causes of AKI (dehydration, sepsis, nephrotoxins).
9. Check for drugs that requiring dosing changes due to renal impairment (e.g. Metformin, Sitagliptin) and adjust dose appropriately.
10. Minimise further damage to kidneys, e.g. treat urinary tract infection, avoid use of NSAIDs.
11. Advise protein diet restriction (Limit to 2 servings of protein daily: 1 serving = 1 palm size (80 to 100 g) protein per main meal).
12. Patients with advanced kidney disease (CKD 3b, 4, 5) should be referred to renal specialist.
13. The presence of albuminuria should alert the physician to the presence of diabetic retinopathy.

Diabetes Retinopathy

Diabetic retinopathy is a major blinding eye disease.

Screening

Patients with T2DM should have a Diabetic Retinal Photography (DRP) screening at the time of diagnosis and at least once yearly subsequently.

Management

1. Manage according to recommendations in DRP report.
2. Good glycaemic control (HbA1c preferably 6.5 to 7.5%) should be instituted to reduce the risk of retinopathy.
3. Rapid normalisation of blood glucose may worsen retinopathy and thus retinal assessment should be carried out before initiation of intensive insulin therapy and then at 3-monthly intervals for 6 - 12 months. Patients should be carefully monitored during this period.
4. Good control of blood pressure at or below 130/80 mmHg should be instituted to reduce the progression of diabetic retinopathy.
5. Significant hyperlipidaemia should be treated to retard diabetic retinopathy. Consideration can be given for the use of fenofibrate in patients with significant retinopathy as it has been shown as part of tertiary outcome analysis in two trials to reduce progression of diabetic retinopathy.
6. Concurrent treatment of anaemia may possibly slow the progression of diabetic retinopathy.

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of diabetes mellitus, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Introduce and enrol patient onto Teleconsult
- Perform general screening (i.e. fall risk, social economics, smoking & drinking history)
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal), cancer screenings (FIT kit, mammogram, PAP smear / HPV), DRP and DFS
- Teach home glucose monitoring to patient

3. Care Manager

- Patient education on diabetes, HbA1c and glucose targets and lifestyle measures
- Teach home glucose monitoring, insulin injection technique / titration while assessing for treatment adherence
- Patients who are stable and have achieved good glycaemic, BP or lipid level control may have alternate clinics visits between the doctor and the care manager

4. Advanced Practice Nurse

- Manage patients with diabetes, hypertension and dyslipidaemia within scope of practice
- Initiate and titrate glucose-lowering medications and insulin therapy
- Manage common acute conditions such as upper respiratory tract infection, gastroenteritis should the patient concomitantly present with these reasons for encounter

5. Clinical Pharmacist

- Manage patients with diabetes, hypertension and dyslipidaemia
- Initiate and titrate glucose-lowering medications
- For patients who need drug optimisation or closer monitoring due to increased risk for drug interactions, polypharmacy, lack of drug-related knowledge or drug non-adherence

6. Dietitian

- Dietary management for patients with Diabetes – meal planning
- Hypoglycaemia management
- Patient education on weight management

7. Podiatrist

- Intervention for foot problems, especially if the patient has diabetic foot ulcer.
- Patient education on prevention and management of diabetic foot complications.

8. Medical Social Worker

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resources and caregiver support

9. Financial Counsellor

- Financial counselling for patients with issues on payment of medical bills
- Advised on MediSave utilisation for chronic disease medical bills
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high cost drugs
- Application on Institutional Medical Fund-Remission for eligible non-residents

10. Psychologist

- Emotional and psychological support to facilitate adjustment to diabetes, and manage diabetes distress.
- Psychological and behavioural interventions to develop coping strategies to cope with the development and progression of diabetes-related complications, improve disease management and quality of life.
- Psychological assessment and intervention to manage co-occurring psychological problems (depression, anxiety disorders etc.)
- Psychological support for carers experiencing caregivers stress

11. Physiotherapist

- Exercise prescription on resistance and aerobic exercises.
- Patient education regarding sedentary periods and physical activity.
- Patient education regarding hypoglycaemia and exercise.

12. Pharmacist

- Smoking cessation clinic
- Detect, prevent and address drug-drug/drug-disease interactions
- Perform medication reconciliation.

PREVENTION OF TYPE 2 DIABETES MELLITUS AND MANAGING PRE-DIABETES

Defining Pre-Diabetes

Pre-diabetes is defined by glycaemic levels that are higher than normal, but lower than diabetes thresholds.

Table 15. Glucose Thresholds for Pre-Diabetes

Pre-diabetes	Fasting plasma glucose (mmol/L)	2-hr post-load glucose (mmol/L)[†]
Impaired fasting glucose (IFG)	6.1 - 6.9	< 7.8
Impaired glucose tolerance (IGT)	< 7.0	7.8 - 11.0

[†]2-hour 75-g oral glucose tolerance test (OGTT)

Refer to [Table 1](#) for interpretation and action for screening of diabetes using HbA1c and follow-up test results

Lifestyle Intervention

Lifestyle intervention is recommended to all persons with pre-diabetes. Refer all patients with pre-diabetes to a dietitian. For patients with a BMI of 23 and above, consider additional referral to a weight management program (i.e. [NU-LITE](#)).

Diet

1. Use My Healthy Plate as a guide to have healthy balanced meals.
2. Regular meal timing: Eating at regular times can help control blood sugar levels. Aim to eat every four to five hours.
3. Calorie reduction: For overweight or obese individuals, reducing energy intake can lead to moderate weight loss, which can significantly improve insulin sensitivity.
4. Carbohydrate management:
 - Emphasise complex carbohydrates such as whole grains and brown rice, oats as they have more fibre vitamins and minerals.
 - Limit refined sugars found in regular soft drinks, fruit juice, sweets, cakes, desserts. Choose water or zero-calorie beverages.
 - Focus on low glycaemic index foods such as milk, vegetables, fruits, nuts, and seeds
 - Control portion sizes of carbohydrate foods at each meal or snack
5. Increase fibre: Foods high in fibre is digested and absorbed slower. This helps to control blood sugar levels. Examples include whole grains, legumes, fruits, and vegetables.
6. Protein intake: Include lean protein sources in meals to help with satiety and blood sugar control.

7. Reduce fat: Limit fat intake and choose unsaturated fats like olive oil, nuts, and fatty fish.

Weight Management

1. A BMI of $< 23 \text{ kg/m}^2$ and/or a 5 to 10% body weight loss should be gradually achieved and maintained by implementing a negative caloric balance.
2. Refer all patients with prediabetes and $\text{BMI} \geq 23 \text{ kg/m}^2$ to a weight management or lifestyle programme and dietitian.
3. Refer all patients with prediabetes and $\text{BMI} < 23 \text{ kg/m}^2$ to dietitian for diet and lifestyle advice.

Physical Activity

Advise persons with pre-diabetes to:

1. Perform at least 150 minutes of moderate-intensity aerobic exercise per week, such as brisk walking, swimming, or cycling.
2. Engage in resistance exercises 2-3 times a week, with at least 1 day of rest between sessions. These can include bodyweight exercises (e.g., push-ups, squats), weightlifting, and resistance band exercises.
3. Consider increasing daily walking by, for example, getting off the bus one stop earlier or parking farther from their destination.

Smoking

1. Individuals, especially those with prediabetes, are strongly advised to quit smoking. Smoking impairs glucose metabolism, insulin sensitivity, and insulin secretion, potentially accelerating the progression from prediabetes to type 2 diabetes.
2. Consider referral to the NUHS Pharmacy-led smoking cessation clinic for nicotine replacement therapy, such as patches, gum, and lozenges. Alternatively, refer patients to smoking cessation services like the HPB Quitline at 1800 438 2000 or the I Quit programme (<https://www.healthhub.sg/programmes/iQuit>).

Pharmacological Therapy

1. Clinicians are encouraged to consider starting metformin in patients if HbA1c is persistently raised $> 6.5\%$ despite 6 months of lifestyle intervention, especially for patients < 60 years old and $\text{BMI} \geq 23 \text{ kg/m}^2$. For older patients, initiation of metformin can be individualised according to each patient and clinical judgement.
2. Measure renal function before initiation of metformin.

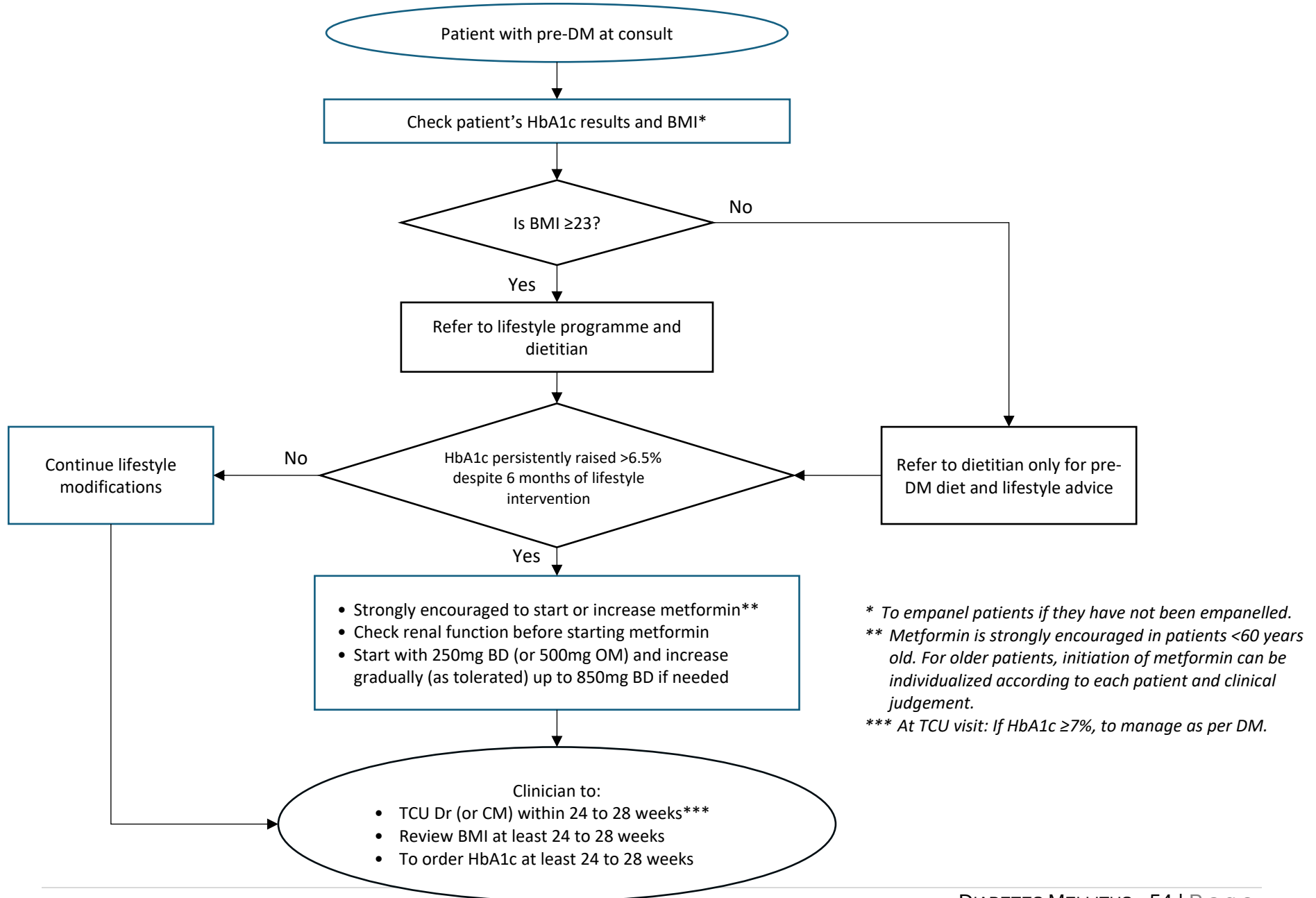
3. Start at 250mg twice daily or 500mg once daily, and gradually increase as tolerated up to 850 mg twice daily, if required.
4. Take metformin with meals to reduce side effects such as nausea, vomiting and diarrhoea.
5. Acarbose has shown a favourable trend in preventing or delaying T2DM in pre-diabetes. However, evidence is not as robust or well-studied as metformin.
6. Consider acarbose only when metformin is not tolerable as it is more costly.
7. Acarbose acts mainly by decreasing postprandial glucose. Hence, its glucose-lowering effect is more likely to benefit persons with IGT and not IFG alone.
8. Common side effects of acarbose include gastrointestinal issues like flatulence, abdominal discomfort, and diarrhoea. These side effects are often most pronounced at the beginning of treatment and may decrease over time.

Follow-Up

1. Follow-up on patient at least once every 24 to 28 weeks with blood pressure, BMI and HbA1c measurement.
2. Monitor fasting venous glucose and lipid panel yearly, unless there are other clinical indications requiring earlier monitoring.
3. Aim to keep HbA1c <6.5%*.
4. If HbA1c $\geq 7\%$, to manage as per type 2 DM.
5. For patients on metformin, perform yearly renal function test unless there is evidence of renal impairment.
6. Periodically measure vitamin B12 levels for persons with established anaemia or peripheral neuropathy, especially if they are on higher doses of metformin.

* A person with HbA1c $\geq 6.5\%$ may have progressed to type 2 DM. Though not routinely done, the clinician may consider performing OGTT 1 to 2 weeks after discontinuing drug therapy. If T2DM is confirmed, proceed to manage as per type 2 DM.

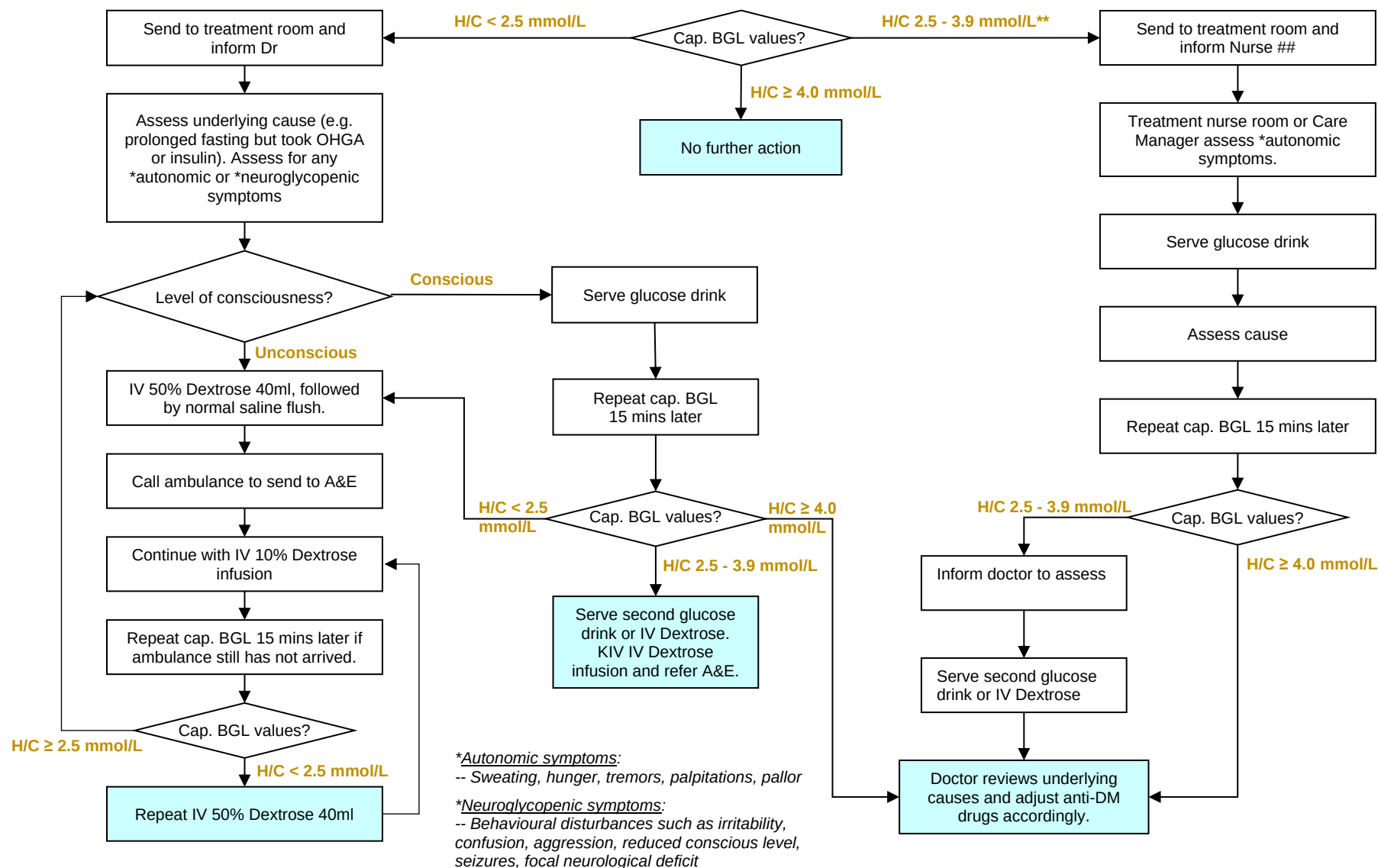
NUP Pre-Diabetes Consult Workflow



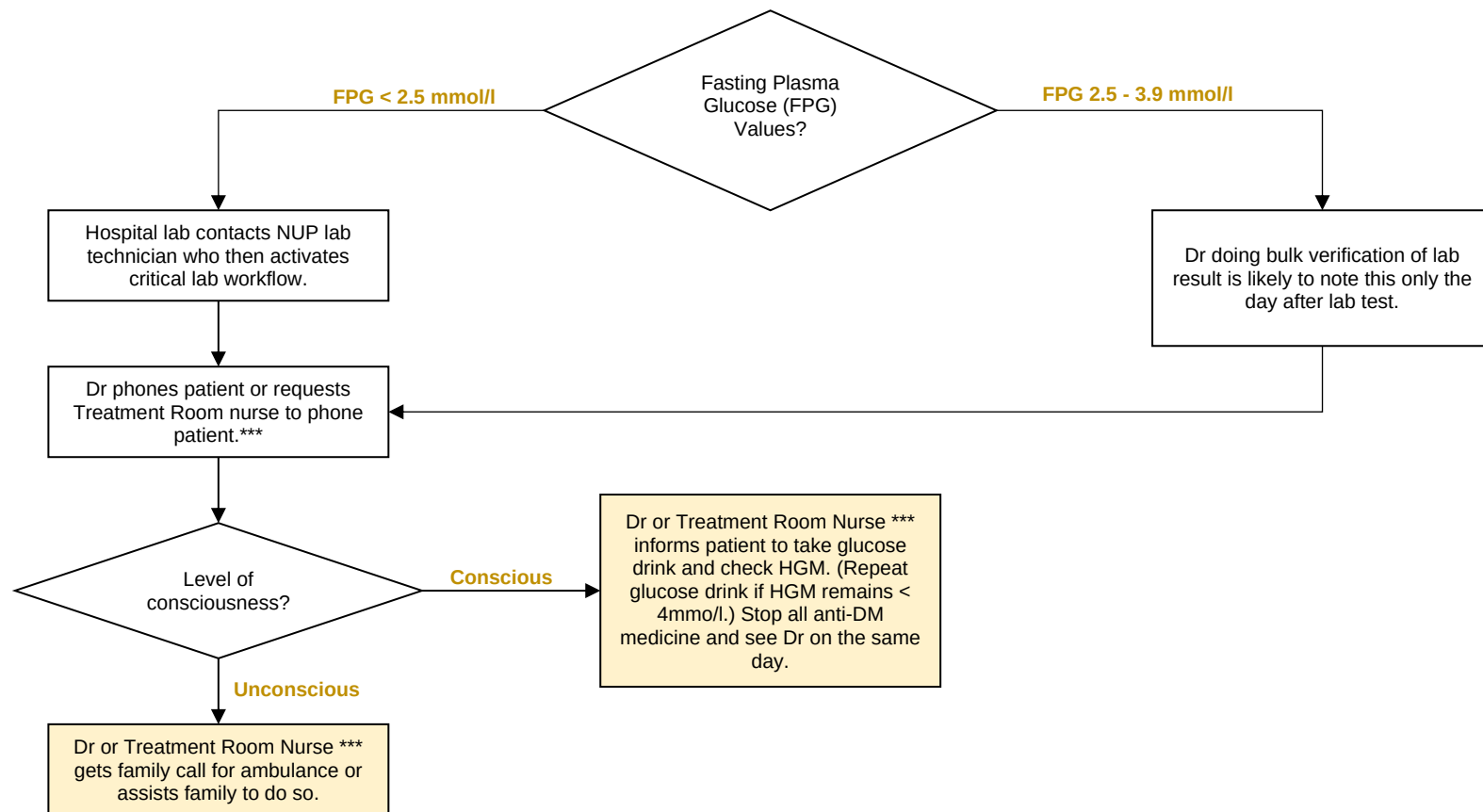
SPECIAL SITUATIONS AND CONSIDERATIONS

Acute Hypoglycaemia and Management

Workflow for patient with hypoglycaemia if he / she is present at the clinic



Workflow for patient with hypoglycaemia if he / she is not present at the clinic



***NB

The workflow involving Treatment Room Nurse takes place only during office hours. After office hours, the Dr covering Critical Lab Workflow will follow up with action plans.

H/C: Hypocount -- synonymous with capillary blood glucose level (Cap. BGL) in this workflow.

HGM: Home Glucose Monitoring

Patient may be sent to care manager's room if the laboratory is next to or near the care manager's room at the same level of the polyclinic.

Some Important Tips for The Doctors on Hypoglycaemia Management

Assess for hypoglycaemia at every visit and identify the precipitating cause. Do not ascribe every outlier to over medication.

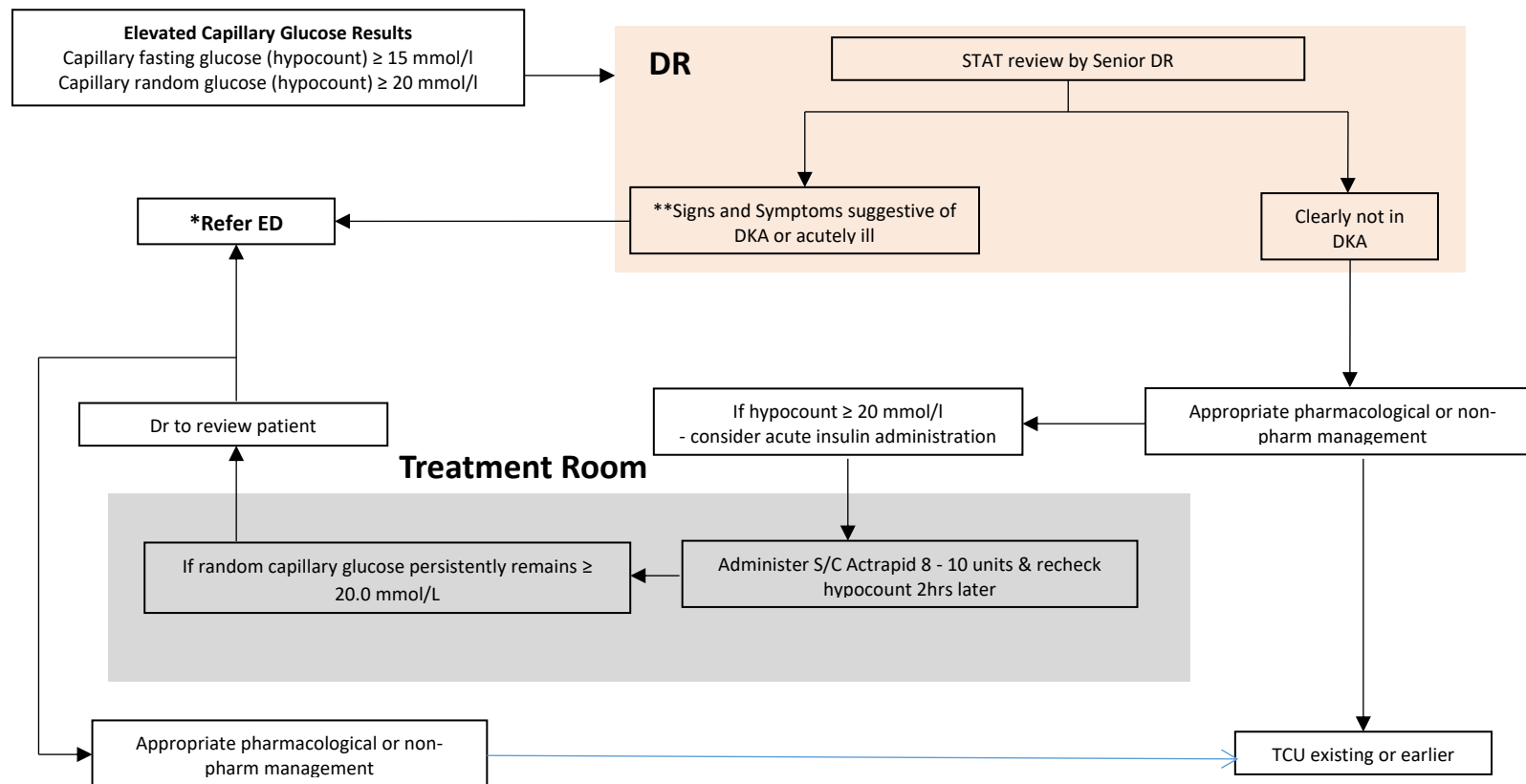
Common Causes of Hypoglycaemia
• Over fasting for laboratory test • Delayed meals • Carbohydrate-insulin mismatch • Acute illness • Alcohol consumption • Intense exercise

Insulin-treated patients with clinically significant hypoglycaemia and hypoglycaemia unawareness should have their glycaemic targets relaxed to avoid hypoglycaemia for several weeks in order to partially reverse hypoglycaemia unawareness.

Acute Hyperglycaemia and Management

Patients with critical capillary glucose (hypocount) values may need to be referred to the Emergency Department. Refer to the following workflow.

Figure 3. Algorithm for the Management of Acute Hyperglycaemia



* For patients with DKA, need to run in 500 ml NS fast and another 500 ml NS over 30 minutes while waiting for ambulance.

** For patients who have other acute illness, to consider ED referral for other acute causes

* Important notes:

- The glucose values refer to capillary (Hypocount) samples.
- The doctor will order the amount of insulin to be administered by the nurse.
- Even if the patient is on a regular insulin regime as part of his usual chronic medication, the treatment room nurse can still proceed and administer Actrapid Insulin as long as his last dose of insulin administered at home is at least an hour earlier.
- Urine dipstick for ketones has poor positive predictive value and clinicians should exercise clinical judgement if opting to use it.

Female Patients at Reproductive Age

All women with diabetes who are in the reproductive age group should receive pre-pregnancy counselling. The aims of pre-pregnancy counselling are:

- to provide education and counselling about the risk of congenital malformation associated with unplanned pregnancy and poor glycaemic control
- to assess suitability for pregnancy
- to provide information about the use of effective contraception, unless the patient is in good metabolic control and trying to conceive
- to adjust and convert drugs like oral glucose-lowering drugs, some antihypertensive medications, and other treatments used pre-pregnancy to alternatives which are safer during pregnancy
- to look for complications of diabetes (especially retinopathy and nephropathy) and to evaluate and treat these complications prior to the onset of pregnancy
- to achieve optimal glycaemic control prior to and during very early pregnancy
- to provide information about what to expect in pregnancy and general measures to improve outcome
- to provide an opportunity for pre-pregnancy dietary advice and folate supplementation

Diabetic patients who become pregnant should be referred to an obstetric SOC for antenatal management as soon as pregnancy is confirmed.

Gestational and Pre-Gestational Diabetes

Diabetes mellitus in pregnancy falls into 2 categories: Gestational diabetes mellitus (GDM) is defined as any degree of glucose intolerance with onset or first recognition during pregnancy. The definition applies whether or not the condition persists after pregnancy. Pre-gestational diabetes is diagnosed when the woman has diabetes before the onset of pregnancy.

The International Association of Diabetes and Pregnant Study Group (IADPSG) 3-point diagnostic criteria for GDM using the 75g OGTT criteria is used for the diagnosis of GDM.

GDM is diagnosed if any of the 3-point diagnostic criteria is met:

- Fasting venous plasma glucose ≥ 5.1 mmol/L
- 1-hour venous plasma glucose ≥ 10.0 mmol/L
- 2-hour venous plasma glucose ≥ 8.5 mmol/L

Patients with pre-gestational diabetes or GDM should be referred to the obstetric SOC for antenatal management.

For women with GDM, OGTT should be performed 6 - 12 weeks postpartum and the patient reclassified according to criteria accepted in the non-pregnant state. Women with a history of gestational diabetes should have lifelong screening for the development of prediabetes or diabetes annually.

Children and Adolescents with DM

Children and adolescents with DM should be referred to the Diabetes Clinic / Centre for management by a multi-disciplinary team.

Patients with Type 1 DM

All patients with suspected T1DM should be referred to the specialist promptly for assessment.

Patients Fasting During Ramadan

(see <https://nupintranet/DOC/Shared%20Documents/CPG/Prescribing%20Guidelines%20for%20Ramadan.docx>)

REFERRALS

1. All pregnant women diagnosed with pre-gestational diabetes and those diagnosed with gestational diabetes should receive specialised care. Consider referring women planning for pregnancy who require pre-conception intensive glucose control.
2. Children and adolescents with suspected diabetes should be referred to a specialist for early assessment, where possible, on the same day.
3. Patients who are suspected to have T1DM or monogenic diabetes should be referred to the specialist promptly for assessment.
4. Consider referral to specialist if patient's insulin requirement is > 1 unit/kg ideal body weight.
5. Consider referral to renal specialist when there is uncertainty about the aetiology of kidney disease (heavy proteinuria, active urine sediment, absence of retinopathy, rapid decline in GFR), difficult management issues, or advanced kidney disease.
6. Patients with advanced kidney disease (CKD 3b, 4, 5) should be referred to renal specialist.
7. Diabetic patients found to have diabetic macular oedema or moderate and more severe non-retinopathy should be referred for further ophthalmological assessment.
8. Urgent referral to a specialised footcare team is needed in the presence of ulcerations, severe foot infection and gangrene.
9. Patients with morbid obesity who are open to the option of intensive weight management including bariatric surgery.

RECOMMENDED CARE COMPONENTS FOR DIABETES MELLITUS

Recommended Care Components	Minimum Frequency*	Remarks
BP measurement~	Twice a year	For those with hypertension, an acceptable treatment-initiation and target blood pressure is generally < 140/80mmHg; Clinicians may personalise the targets accordingly based on patients' risk factors
Weight and BMI measurement~	Twice a year	Keep < 23kg/m ² (For non-Asian population, keep BMI < 25 kg/m ²)
Lipid profile~	Annually	All patients should be risk stratified (as recommended in the Lipids CPG) Targets of treatment should be personalised by levels of risk
Glycated Haemoglobin (HbA1c)~	Twice a year	General HbA1c target of ≤ 7.0%, but target of treatment should be personalised (e.g. for elderly)
Kidney Assessment~ (i) Serum Cr and/or eGFR, and (ii) Urine Albumin-Creatinine Ratio (uACR) or Protein-Creatinine Ratio (uPCR)	Annually	Good glycaemic control and good BP control with ACE-I or ARB preferred to slow progression of diabetic nephropathy Annual screening of a) serum Cr and/or eGFR and b) uACR in all patients, or uPCR if significant levels of proteinuria (as defined in evidence-based practice guidelines)
Eye Assessment~	Annually	Includes retinal photography and visual acuity <u>Patients with T1 DM</u> : First assessment within 3-5 years after diagnosis of diabetes once patient is aged ten years or older, then annually

Recommended Care Components	Minimum Frequency*	Remarks
		<u>Patients with T2 DM</u> : First assessment at diagnosis, then annually
Foot Assessment~	Annually	Screen for peripheral neuropathy, peripheral vascular disease, bone, joint, skin and nail abnormalities, and poor footwear
Smoking assessment~	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Cardiac assessment	At diagnosis before initiating medications	Includes baseline ECG
Influenza Vaccination#	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS) and National Childhood Immunisation Schedule (NCIS)
Pneumococcal Vaccination (PCV 13 ± PPSV23 or PCV20) #	1 or 2 doses depending on vaccine type, age and other medical conditions	As recommended under the NAIS and NCIS (Refer to <u>NUP detailed recommendation on pneumococcal vaccines</u>)
Shingles Vaccination (Recombinant herpes zoster vaccine)	2 doses at an interval of 2 to 6 months	As recommended under the NAIS and NCIS (patients aged 60 years or older)

~ These are reportable clinical indicators

*More frequently if clinically indicated

These are non-reportable clinical indicators

ABBREVIATIONS

<	less than
≤	less than or equal to
>	more than
≥	more than or equal to
1 st	first
24h CCT	24-hour urinary creatinine clearance test
24h UTP	24-hour urinary total protein
2HPP	2-hour postprandial (capillary prick)
b.w	bodyweight
ABI	Ankle-Brachial Index
ACE-I	Angiotensin-Converting Enzyme Inhibitor
ARB	Angiotensin II receptor antagonist
asym.	asymptomatic
BGL	blood glucose level
BMI	body mass index
Cr	creatinine
CVD	cardiovascular disease
DFU	diabetic foot ulcer
DKA	diabetic ketoacidosis
DM	diabetic mellitus
DRP	diabetic retinal photography
DFS	diabetic foot screening
Dx	diagnosis
ECG	electrocardiogram
eGFR	estimated Glomerular Filtration Rate
FBG	fasting blood glucose (capillary prick)
FPG	fasting plasma glucose (venous)
GDM	Gestational Diabetes
GN	glomerulonephritis
HbA _{1c}	glycosylated haemoglobin
HDLc	high density lipoprotein cholesterol
HHS	hyperglycaemic hyperosmolar syndrome
IFG	impaired fasting glycaemia
IGT	impaired glucose tolerance
IUCD	intrauterine contraceptive device
LDLc	low density lipoprotein cholesterol

LEA	lower extremity amputation
MSW	Medical Social Worker
OGTT	oral glucose tolerance test
PAD	peripheral arterial disease
PVD	peripheral vascular disease
prn	when necessary
RBG	random blood glucose (capillary)
RPG	random plasma glucose
SMBG	Self-Monitoring of Blood Glucose
SOC	Specialist Outpatient Clinic
sym.	symptomatic
Tc	total cholesterol
TG	triglyceride
UFEME	urine full examination and microscopic examination
UACR	urine albumin-creatinine ratio
UPCR	urine protein-creatinine ratio
U/S kidneys	ultrasonic examination of kidneys
UTI	urinary tract infection
yr	year

Note: DM panel test consists of serum electrolytes, creatinine, glucose, HbA_{1c}, lipid profile, urine albumin-creatinine ratio.

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